Double redshift for localizing motives

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Abstract

Let R be a commutative ring spectrum of finite nonnegative telescopic height at a prime p. We prove that the continuous algebraic K-theory of the category of localizing motives over R has telescopic height $\mathrm{ht}_p(R)+2$. The upper bound follows from the purity theorem for K-theory applied twice. For the lower bound we find one ring at each height whose motives are detected by an iterated trace, and move this nonvanishing to every ring of the same height. This uses two properties of continuous K-theory of motives as a function of the coefficients: it only sees the coefficients near the relevant heights, and its Bousfield class does not change under finite faithfully étale extension. The second property is proved with multiplicative norms on equivariant localizing motives.

1 Introduction

Algebraic K-theory raises chromatic height by one: if a commutative ring spectrum R has telescopic height n, then $K(R)$ has height $n+1$. The upper bound is the purity theorem of Land, Mathew, Meier and Tamme [14], and the lower bound follows from the chromatic Nullstellensatz of Burklund, Schlank and Yuan [3]. We prove the analogous statement one categorical level up, with $K(R)$ replaced by the K-theory of the category of localizing motives over R.

Fix a prime p.

Definition 1.1 (Telescopic height)**.** For $j \geq 1$, let $T(j)$ be the mapping telescope of a v_j-self-map of a finite p-local spectrum of type j, and put $T(0) = H\mathbb{Q}$. The *telescopic height* of a spectrum X at p is

$$\mathrm{ht}_p(X) = \sup\{j \geq 0 \mid L_{T(j)}X \neq 0\},$$

with the supremum of the empty set taken to be -1.

Definition 1.2 (Localizing motives)**.** Let R be a commutative ring spectrum, and let $\mathrm{Cat}_R^{\mathrm{perf}}$ be the category of small idempotent-complete R-linear stable categories and exact R-linear functors. A *Karoubi sequence* $\mathcal{A} \to \mathcal{B} \to \mathcal{C}$ is a null composite with $\mathcal{A} \to \mathcal{B}$ fully faithful that identifies $\mathcal{C}$ with the idempotent completion of $\mathcal{B}/\mathcal{A}$. A functor from $\mathrm{Cat}_R^{\mathrm{perf}}$ to a stable category is a *localizing invariant* if it takes Karoubi sequences to cofiber sequences, and *finitary* if it preserves filtered colimits.

The category $\mathrm{Mot}_R^{\mathrm{loc}}$ of *localizing motives over* R is the presentable stable category equipped with the universal finitary localizing invariant

$$U_R : \mathrm{Cat}_R^{\mathrm{perf}} \longrightarrow \mathrm{Mot}_R^{\mathrm{loc}};$$

that is, for every presentable stable $\mathcal{E}$, precomposition with U_R identifies colimit-preserving functors $\mathrm{Mot}_R^{\mathrm{loc}} \to \mathcal{E}$ with finitary localizing invariants $\mathrm{Cat}_R^{\mathrm{perf}} \to \mathcal{E}$.

By [12, Theorem 5.18], applied to Perf_R, the relative tensor product of R-linear categories descends to a presentably symmetric monoidal structure on $\mathrm{Mot}_R^{\mathrm{loc}}$ for which U_R is symmetric monoidal. In particular,

$$U_R\mathcal{C} \otimes U_R\mathcal{D} \simeq U_R(\mathcal{C} \otimes_R \mathcal{D}), \qquad \mathbf{1}_{\mathrm{Mot}_R^{\mathrm{loc}}} = U_R\mathrm{Perf}_R.$$

Efimov proves that $\mathrm{Mot}_R^{\mathrm{loc}}$ is dualizable and rigid [7, Theorem 3.1(i)–(ii)], so its continuous K-theory $K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}})$ is defined. The construction $R \mapsto K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}})$ is due to Daming Zhou.

Since $\mathrm{Mot}_R^{\mathrm{loc}}$ is symmetric monoidal, $K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}})$ is a commutative ring spectrum. The endomorphism ring of the unit $U_R\mathrm{Perf}_R$ is $K(R)$, so we obtain a symmetric monoidal functor $\mathrm{Mod}_{K(R)} \to \mathrm{Mot}_R^{\mathrm{loc}}$ and hence a ring map

$$K(K(R)) \longrightarrow K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}}).$$

This suggests that $K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}})$ has the height of K-theory applied twice. Rigidity alone does not imply this, since Ramzi constructs rigid symmetric monoidal categories in which the endomorphism ring of the unit is $T(n)$-local and the K-theory has the same height as this ring [18]. Our proof instead uses the universal property of localizing motives under change of coefficients.

Theorem 1.3. *If R is a commutative ring spectrum with $0 \le \mathrm{ht}_p(R) < \infty$, then*

$$\mathrm{ht}_p(K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}})) = \mathrm{ht}_p(R) + 2.$$

The upper bound is formal. Applying purity to R and then to $K(R)$, we see that $K(K(R))$ vanishes above height $\mathrm{ht}_p(R) + 2$. Since there is a ring map $K(K(R)) \to K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}})$, the same holds for $K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}})$ (Proposition 2.4).

The lower bound is the substance of the paper. If $R \to R'$ is a ring map and a localization of $K^{\mathrm{cont}}(\mathrm{Mot}_{R'}^{\mathrm{loc}})$ is nonzero, then the same localization of $K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}})$ is nonzero (Lemma 2.3). By the chromatic Nullstellensatz, every R of height $n \ge 1$ admits a ring map to the Lubin–Tate theory $E_n(k, \Gamma_0)$ of a formal group Γ_0 of height n over an algebraically closed field k of characteristic p. Every R of height 0 admits a ring map to $Hk[u^{\pm 1}]$ for an algebraically closed field k of characteristic zero. It therefore suffices to show that $L_{T(n+2)}K^{\mathrm{cont}}(\mathrm{Mot}_{R'}^{\mathrm{loc}}) \neq 0$ for these targets R'.

We exhibit a single ring C of height n with $L_{T(n+2)}K^{\mathrm{cont}}(\mathrm{Mot}_C^{\mathrm{loc}}) \neq 0$. To transport this nonvanishing to the targets R', we prove two properties of $R \mapsto K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}})$.

(i) *Purity.* For $j \ge 2$, a map $R \to S$ of p-local commutative ring spectra that is a $T(j-2) \vee T(j-1) \vee T(j)$-equivalence induces an equivalence $L_{T(j)}K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}}) \simeq L_{T(j)}K^{\mathrm{cont}}(\mathrm{Mot}_S^{\mathrm{loc}})$.

(ii) *Finite étale invariance.* If S/R is finite faithfully étale, then $K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}})$ and $K^{\mathrm{cont}}(\mathrm{Mot}_S^{\mathrm{loc}})$ have the same Bousfield class: a spectrum E satisfies $E \otimes K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}}) = 0$ if and only if $E \otimes K^{\mathrm{cont}}(\mathrm{Mot}_S^{\mathrm{loc}}) = 0$.

By (i), we may invert a periodicity class in C and then $K(n)$-localize, since both maps are $T(n) \vee T(n+1) \vee T(n+2)$-equivalences. By Lazard's theorem, the isomorphisms between the formal group of C and Γ_0 are classified by an ind-finite étale algebra. Property (ii), extended to filtered colimits, therefore allows us to adjoin these isomorphisms. By the universal property of Lubin–Tate theory, we then obtain a ring map from $E_n(k, \Gamma_0)$ to the resulting extension of C. At height zero, we use (ii) to reduce the comparison between fields of characteristic zero to finite separable extensions.

We obtain the detecting ring C from a Lubin–Tate theory of height $n+2$ by applying the construction $X \mapsto (\tau_{\geq 0}X)^{tC_p}$ twice, with C_p acting trivially. By blueshift, the height of C is at most n. On the other hand, the iterated trace

$$K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}}) \longrightarrow \mathrm{THH}(\mathrm{THH}(R))^{h(S^1\times S^1)},$$

applied to $R = C$, still detects height $n+2$ (Proposition 3.3).

Property (i) is Corollary 2.7, and we deduce it from ordinary purity. Efimov's self-duality of $\mathrm{Mot}_R^{\mathrm{loc}}$ pairs the motives of $\mathcal{C}$ and $\mathcal{D}$ to $K(\mathcal{C}\otimes_R \mathcal{D})$. Hence ordinary purity [14] applies to motives, and Jin's purity theorem for dualizable categories [13] transfers the result to continuous K-theory.

Property (ii) is Theorem 2.10, proved in the second half of the paper. One implication follows from the ring map $K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}}) \to K^{\mathrm{cont}}(\mathrm{Mot}_S^{\mathrm{loc}})$. For the other, let S/R be a G-torsor. It suffices to factor the unit of $K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}})$ through a spectrum that is E-acyclic whenever $K^{\mathrm{cont}}(\mathrm{Mot}_S^{\mathrm{loc}})$ is. We use the G-fixed points of $\mathrm{N}_e^G K^{\mathrm{cont}}(\mathrm{Mot}_S^{\mathrm{loc}})$, through which we factor the unit by a multiplicative transfer from S to R. Its construction has two steps.

First, we show that the categories $\mathrm{Mot}_{S^{hH}}^{\mathrm{loc}}$ form a genuine G-category $\underline{\mathrm{Mot}}_{S/R}^{\mathrm{loc}}$, and that for $H \leq K \leq G$, finite étale tensor induction of linear categories induces a norm $\mathrm{Mot}_{S^{hH}}^{\mathrm{loc}} \to \mathrm{Mot}_{S^{hK}}^{\mathrm{loc}}$ (Theorem 6.9). The main point is that tensor induction preserves motivic equivalences. We prove this by filtering the norm of a Karoubi sequence and applying isotropy separation. Second, we need continuous K-theory to turn norms of dualizable categories into norms of spectra. Hilman and Ramzi construct such pairings for compactly generated categories [10, Corollary 3.3.3], and we extend them to dualizable genuine categories by means of a Calkin resolution (Theorem 5.6).

In Section 2 we prove the upper bound and property (i), and we state property (ii). In Section 3 we construct the detecting rings and prove the theorem at height zero, and in Section 4 at positive height. In Section 5 we construct the continuous norm pairings, and in Section 6 we construct the norms on relative motives and prove property (ii). Appendix A recalls indexed tensor products of genuine presentable categories in the Sp^G-module model.

Conventions. All categories are stable unless otherwise stated, and all ring maps preserve units. We write $\mathrm{Pr}^L_{\mathrm{st}}$ for the category of presentable stable categories and colimit-preserving functors. Tensor products of presentable categories are taken in $\mathrm{Pr}^L_{\mathrm{st}}$, and a subscript denotes a relative tensor product. Dualizability and rigidity are recalled in Conventions 2.1. Ordinary K-theory is nonconnective, and K^{cont} is the continuous extension of Definition 2.2; in particular, $K^{\mathrm{cont}}(\mathrm{Mod}_R) = K(R)$. A *unital map* of ring spectra is a map of underlying spectra compatible with the units. For a field k of characteristic zero, $Hk[u^{\pm 1}]$ denotes the commutative spectral Laurent algebra with $|u| = 2$. For $t \geq 0$, we write L_t for chromatic $E(t)$-localization and $K(t)$ for Morava K-theory at p.

2 Coefficient change and the upper bound

Conventions 2.1 (Dualizability and rigidity). A presentable stable category is *dualizable* if it is dualizable in $\mathrm{Pr}^L_{\mathrm{st}}$, equivalently a retract in $\mathrm{Pr}^L_{\mathrm{st}}$ of a compactly generated category [5]. A functor is *strongly continuous* if it preserves colimits and so does its right adjoint. A presentable stable monoidal category is *rigid* if its tensor unit is compact and its multiplication has a colimit-preserving bimodule-linear right adjoint [7, Section 1.2]. A presentably symmetric monoidal stable category $\mathcal{A}$ is *locally rigid* if it is dualizable in $\mathrm{Pr}^L_{\mathrm{st}}$ and its multiplication $\mathcal{A}\otimes\mathcal{A} \to \mathcal{A}$ has a colimit-preserving $\mathcal{A}$-bimodule-linear right adjoint [17, Definition 4.5, with $\mathcal{V} = \mathrm{Sp}$].

To define K-theory of dualizable categories, we use Efimov's extension theorem [6, Theorem 1.19]: every accessible localizing invariant of compactly generated categories extends uniquely to dualizable categories and strongly continuous functors.

Definition 2.2 (Continuous K-theory)**.** Write $\mathrm{Cat}_{\mathrm{st}}^{\mathrm{dual}}$ for the category of dualizable presentable stable categories and strongly continuous functors. *Continuous K-theory* is the accessible localizing extension of nonconnective K-theory

$$K^{\mathrm{cont}} : \mathrm{Cat}_{\mathrm{st}}^{\mathrm{dual}} \longrightarrow \mathrm{Sp}$$

provided by Efimov's extension theorem. It is the unique such extension equipped with the natural identification

$$K^{\mathrm{cont}}(\mathrm{Ind}\,\mathcal{C}) \simeq K(\mathcal{C})$$

for small idempotent-complete stable categories $\mathcal{C}$ and exact functors between them.

Continuous K-theory is lax symmetric monoidal. To see this, recall that the universal continuous localizing invariant is symmetric monoidal [5, Remark 4.15], and that the continuous extension is the right Kan extension along Ind [5, Theorem 4.16]. Hence K^{cont} takes commutative algebras in $\mathrm{Cat}_{\mathrm{st}}^{\mathrm{dual}}$ to commutative ring spectra.

2.1 Multiplicative functoriality

Lemma 2.3 (Unit detection)**.** *Let B be a unital ring spectrum and Z a spectrum. If the unit of B factors through a spectrum X with $Z \otimes X = 0$, then $Z \otimes B = 0$. In particular, a unital map $A \to B$ carries Z-acyclicity from A to B.*

Proof. Since $Z \otimes X = 0$, the factorization shows that $Z \otimes \eta_B$ is null. If we tensor it with B and compose with the multiplication of B, we obtain the identity of $Z \otimes B$. Hence this identity is null, and $Z \otimes B = 0$. For the last assertion, take $X = A$. □

To bound the height of $K(K(R))$, we use ordinary chromatic purity [14, Purity Theorem] in the following form. Let $j \geq 1$, and let $A \to B$ be a $T(j-1) \vee T(j)$-equivalence of $\mathbb{E}_1$-rings, where for $j = 1$ we assume that A and B are p-local. Then the induced map $K(A) \to K(B)$ is a $T(j)$-equivalence. We need the restriction at $j = 1$ because [14, Section 2.1] takes $T(0) = \mathbb{S}[1/p]$, and this agrees with our $T(0) = H\mathbb{Q}$ only on p-local spectra.

Proposition 2.4 (Upper bound)**.** *For every commutative ring spectrum R of finite nonnegative telescopic height,*

$$\mathrm{ht}_p(K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}})) \leq \mathrm{ht}_p(R) + 2.$$

Proof. The tensor unit $U_R\mathrm{Perf}_R$ is compact by rigidity [7, Theorem 3.1(ii)], and its endomorphism ring is $K(R)$ by corepresentability of nonconnective K-theory [12, Theorem 5.25]. The associated symmetric monoidal functor $\mathrm{Mod}_{K(R)} \to \mathrm{Mot}_R^{\mathrm{loc}}$ is strongly continuous, since its right adjoint is the mapping spectrum out of the compact unit. Applying continuous K-theory, we obtain a unital map $K(K(R)) \to K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}})$.

Let $m > \mathrm{ht}_p(R) + 2$. Then R is acyclic at the positive heights $m-2$, $m-1$ and m. By purity, $K(R)$ is acyclic at heights $m-1$ and m, and applying purity again we find that $K(K(R))$ is acyclic at height m. By Lemma 2.3, so is $K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}})$, which proves the bound. □

2.2 Coefficient purity

By Efimov's self-duality, a motive is detected by its pairings with motives of linear categories, which are ordinary K-theory spectra. Applying ordinary purity to them, we obtain coefficient purity for motives, and Jin's theorem transfers it to continuous K-theory.

Lemma 2.5 (Self-duality and detection)**.** *There is a perfect pairing*

$$\mathrm{ev} : \mathrm{Mot}_R^{\mathrm{loc}} \otimes \mathrm{Mot}_R^{\mathrm{loc}} \longrightarrow \mathrm{Sp}$$

such that $\mathrm{ev}(U_R\mathcal{C}, U_R\mathcal{D}) \simeq K(\mathcal{C} \otimes_R \mathcal{D})$ *for small* R*-linear categories* $\mathcal{C}, \mathcal{D}$*. Let* E *be a spectrum, and let* Sp_E *be the symmetric monoidal category of* E*-local spectra. An object* N *of* $\mathrm{Sp}_E \otimes \mathrm{Mot}_R^{\mathrm{loc}}$ *is zero if and only if* $\mathrm{ev}(U_R\mathcal{C}, N)$ *is zero in* Sp_E *for every small* R*-linear category* $\mathcal{C}$*, where* ev *also denotes the* Sp_E*-linear extension of the pairing.*

Proof. Efimov's theorem [7, Theorem 3.1(i)], applied to the rigid base Mod_R, identifies the dual of $\mathrm{Mot}_R^{\mathrm{loc}}$ with motives over Mod_R with reversed multiplication; since Mod_R is symmetric monoidal, this is $\mathrm{Mot}_R^{\mathrm{loc}}$ itself. On motives of dualizable R-linear categories, the evaluation is the continuous K-theory of their relative tensor product. For Ind completions of small categories, this is the K-theory in the statement.

After extension of scalars to Sp_E, the pairing identifies an object N of $\mathrm{Sp}_E \otimes \mathrm{Mot}_R^{\mathrm{loc}}$ with the colimit-preserving Sp_E-linear functor $\mathrm{ev}(-, N)$. Since the motives $U_R\mathcal{C}$ generate $\mathrm{Mot}_R^{\mathrm{loc}}$ under colimits, this functor is zero if and only if it vanishes on them. □

Proposition 2.6 (Purity of localizing motives)**.** *Let* $R \to S$ *be a map of* p*-local commutative ring spectra, and let* J *be a nonempty finite set of positive integers. Suppose that* $R \to S$ *is a* $T(j-1) \vee T(j)$*-equivalence for every* $j \in J$*. Then base change induces an equivalence*

$$\mathrm{Sp}_{\bigvee_{j\in J} T(j)} \otimes \mathrm{Mot}_R^{\mathrm{loc}} \xrightarrow{\sim} \mathrm{Sp}_{\bigvee_{j\in J} T(j)} \otimes \mathrm{Mot}_S^{\mathrm{loc}}.$$

Proof. Base change has a continuous right adjoint [7, Proposition 5.1]; write $f : \mathrm{Mot}_R^{\mathrm{loc}} \rightleftarrows \mathrm{Mot}_S^{\mathrm{loc}} : g$, so that

$$f(U_R\mathcal{C}) = U_S(\mathcal{C} \otimes_R S), \qquad g(U_S\mathcal{D}) = U_R(\mathcal{D}|_R).$$

We show that the unit and the counit of the adjunction extended to $\mathrm{Sp}_{\bigvee_{j\in J} T(j)}$ are equivalences. In both cases, we use Lemma 2.5 to reduce the claim to a map of K-theory spectra of linear categories, and ordinary purity shows that this map is a $T(j)$-equivalence.

The unit. Since $\mathrm{Mot}_R^{\mathrm{loc}}$ is rigid, g satisfies the projection formula $g(f(M) \otimes N) \simeq M \otimes g(N)$. Hence the composite gf is given by tensoring with $U_R\mathrm{Perf}_S = g(\mathbf{1})$, and the unit of the adjunction is induced by $\mathbf{1} \to U_R\mathrm{Perf}_S$. Let $N = \mathrm{cofib}(\mathbf{1} \to U_R\mathrm{Perf}_S)$. By Lemma 2.5, we have

$$\mathrm{ev}(U_R\mathcal{C}, N) \simeq \mathrm{cofib}\big(K(\mathcal{C}) \longrightarrow K(\mathcal{C} \otimes_R S)\big).$$

Write $\mathcal{C}$ as the filtered union of its finitely generated thick subcategories, each of the form Perf_A for an $\mathbb{E}_1$ R-algebra A. For every $j \in J$, the map $A \to A \otimes_R S$ is a $T(j-1) \vee T(j)$-equivalence, since it is obtained by applying $A \otimes_R (-)$ to $R \to S$. Ordinary purity [14, Purity Theorem], which allows nonconnective algebras, therefore shows that the displayed cofiber is $T(j)$-acyclic for $\mathcal{C} = \mathrm{Perf}_A$. Passing to the filtered colimit, we obtain the same for $\mathcal{C}$. By Lemma 2.5, the image of N in $\mathrm{Sp}_{\bigvee_{j\in J} T(j)} \otimes \mathrm{Mot}_R^{\mathrm{loc}}$ is zero, and hence the localized unit is an equivalence.

The counit. We pair the value of the counit on $U_S\mathcal{D}$ with $U_S\mathcal{C}$, using Lemma 2.5 for S. Since $fg(U_S\mathcal{D}) = U_S(\mathcal{D}|_R \otimes_R S)$, the resulting comparison map is

$$K(\mathcal{C}|_R \otimes_R \mathcal{D}|_R) \longrightarrow K(\mathcal{C} \otimes_S \mathcal{D}). \tag{2.1}$$

For $\mathcal{C} = \mathrm{Perf}_A$ and $\mathcal{D} = \mathrm{Perf}_B$, it is induced by $A \otimes_R B \to A \otimes_S B$. Comparing bar constructions, we see that this map is a $T(j-1) \vee T(j)$-equivalence, and hence ordinary purity shows that (2.1) is a $T(j)$-equivalence. By filtered generation in each variable, (2.1) is a $T(j)$-equivalence for all $\mathcal{C}$ and $\mathcal{D}$. Lemma 2.5 for S now shows that the localized counit is an equivalence. □

Corollary 2.7 (Chromatic purity)**.** *Let $j \geq 2$. If a map $R \to S$ of p-local commutative ring spectra is a $T(j-2) \vee T(j-1) \vee T(j)$-equivalence, then*

$$L_{T(j)}K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}}) \xrightarrow{\sim} L_{T(j)}K^{\mathrm{cont}}(\mathrm{Mot}_S^{\mathrm{loc}}).$$

Proof. Since $j \geq 2$, we may apply Proposition 2.6 with $J = \{j-1, j\}$, and we obtain

$$\mathrm{Sp}_{T(j-1)\vee T(j)} \otimes \mathrm{Mot}_R^{\mathrm{loc}} \simeq \mathrm{Sp}_{T(j-1)\vee T(j)} \otimes \mathrm{Mot}_S^{\mathrm{loc}}.$$

By Jin's purity theorem, for every dualizable category $\mathcal{C}$ there is a natural equivalence

$$L_{T(j)}K^{\mathrm{cont}}(\mathcal{C}) \simeq L_{T(j)}K^{\mathrm{cont}}(\mathrm{Sp}_{T(j-1)\vee T(j)} \otimes \mathcal{C})$$

[13, Theorem 3.1, using Proposition 2.41]. Using the naturality of this equivalence for the strongly continuous base change $\mathrm{Mot}_R^{\mathrm{loc}} \to \mathrm{Mot}_S^{\mathrm{loc}}$, we obtain the result. □

2.3 Finite étale extension

Definition 2.8 (Finite faithfully étale extensions)**.** A map $R \to S$ of commutative ring spectra is *finite étale* if $\pi_0 S$ is finite étale over $\pi_0 R$ and the natural map

$$\pi_* R \otimes_{\pi_0 R} \pi_0 S \longrightarrow \pi_* S$$

is an isomorphism. It is *finite faithfully étale* if, in addition, $\operatorname{Spec} \pi_0 S \to \operatorname{Spec} \pi_0 R$ is surjective.

Recall that taking π_0 identifies étale commutative R-algebras with étale $\pi_0 R$-algebras, also when R is nonconnective [15, Theorem 7.5.0.6].

Conventions 2.9 (Bousfield classes). The *Bousfield class* $\langle X \rangle$ of a spectrum X is specified by its acyclic spectra: a spectrum E is X-acyclic if $E \otimes X = 0$.

Theorem 2.10 (Finite étale invariance of Bousfield classes)**.** *If $R \to S$ is finite faithfully étale, then*

$$\langle K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}}) \rangle = \langle K^{\mathrm{cont}}(\mathrm{Mot}_S^{\mathrm{loc}}) \rangle.$$

Equivalently, for every spectrum E,

$$E \otimes K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}}) = 0 \quad \Longleftrightarrow \quad E \otimes K^{\mathrm{cont}}(\mathrm{Mot}_S^{\mathrm{loc}}) = 0.$$

In particular, for every $n \geq 0$,

$$L_{T(n)}K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}}) = 0 \quad \Longleftrightarrow \quad L_{T(n)}K^{\mathrm{cont}}(\mathrm{Mot}_S^{\mathrm{loc}}) = 0.$$

Lemma 2.3, applied to the ring map $K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}}) \to K^{\mathrm{cont}}(\mathrm{Mot}_S^{\mathrm{loc}})$, gives the implication from R to S. We prove the converse in Section 6.

Let $S = \operatorname{colim}_i S_i$ be a filtered colimit of finite étale extensions. The next two lemmas show that if $K^{\mathrm{cont}}(\mathrm{Mot}_S^{\mathrm{loc}})$ is $T(n)$-acyclic, then so is some $K^{\mathrm{cont}}(\mathrm{Mot}_{S_i}^{\mathrm{loc}})$.

Lemma 2.11. *The functor $R \mapsto K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}})$ preserves filtered colimits of commutative ring spectra, including the unital multiplicative structures.*

Proof. For motives this is [7, Proposition 5.3], and for continuous K-theory it is [5, Corollary 4.9]. Both comparison maps are multiplicative. □

Lemma 2.12 (Finite witness)**.** *Let $m \geq 0$, and let $B = \operatorname{colim}_i B_i$ be a filtered colimit of unital $\mathbb{E}_1$-rings. If $T(m) \otimes B = 0$, then $T(m) \otimes B_i = 0$ for some i.*

Proof. We may p-localize the diagram. For $m \geq 1$, choose a finite p-local type-m spectrum V with a self-map $v : \Sigma^d V \to V$ defining $T(m)$. The spectrum $T(m) \otimes B$ is the colimit of $V \otimes B \xrightarrow{v} \Sigma^{-d} V \otimes B \xrightarrow{v} \cdots$, and it vanishes. Since V is compact, the map $\mathrm{id} \otimes \eta_B : V \to V \otimes B$ becomes null at a finite stage of this colimit. In other words, some composite

$$\Sigma^{Nd} V \xrightarrow{v^N} V \xrightarrow{\mathrm{id} \otimes \eta_B} V \otimes B$$

is null. By compactness of $\Sigma^{Nd} V$, this nullhomotopy descends to some stage B_i. Its right B_i-linear extension

$$v^N \otimes \mathrm{id}_{B_i} : \Sigma^{Nd} V \otimes B_i \longrightarrow V \otimes B_i$$

is therefore null, and hence its telescope $T(m) \otimes B_i$ vanishes. For $m = 0$, the same argument applies with multiplication by p on the p-local sphere in place of v. □

Corollary 2.13 (Ind-finite-étale support invariance)**.** *Suppose that $S \simeq \operatorname{colim}_i S_i$ is a filtered colimit of commutative R-algebras and each S_i/R is finite faithfully étale. For every $n \geq 0$,*

$$L_{T(n)} K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}}) = 0 \quad \Longleftrightarrow \quad L_{T(n)} K^{\mathrm{cont}}(\mathrm{Mot}_S^{\mathrm{loc}}) = 0.$$

Proof. The implication from R to S follows from Lemma 2.3, applied to the ring map $K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}}) \to K^{\mathrm{cont}}(\mathrm{Mot}_S^{\mathrm{loc}})$. Conversely, suppose that $L_{T(n)} K^{\mathrm{cont}}(\mathrm{Mot}_S^{\mathrm{loc}}) = 0$. By Lemma 2.11, $K^{\mathrm{cont}}(\mathrm{Mot}_S^{\mathrm{loc}})$ is the filtered colimit of the unital rings $K^{\mathrm{cont}}(\mathrm{Mot}_{S_i}^{\mathrm{loc}})$. Hence Lemma 2.12 shows that $L_{T(n)} K^{\mathrm{cont}}(\mathrm{Mot}_{S_i}^{\mathrm{loc}}) = 0$ for some i, and Theorem 2.10, applied to S_i/R, gives $L_{T(n)} K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}}) = 0$. □

3 The double Tate construction and height zero

For $m \geq 2$, we construct from a Lubin–Tate theory of height m a ring whose coefficients vanish above height $m - 2$, but whose motives have nonzero $T(m)$-local continuous K-theory (Proposition 3.3). The nonvanishing is detected by an iterated trace. For $m = 2$, coefficient purity allows us to rationalize this ring, and finite étale invariance then gives the lower bound at height zero.

3.1 The iterated trace

We construct the detecting map by composing the categorified Chern character of Hoyois, Scherotzke and Sibilla with the Dennis trace.

Proposition 3.1 (Iterated trace)**.** *For every commutative ring spectrum R, there is a natural unital map of commutative ring spectra*

$$K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}}) \longrightarrow \mathrm{THH}(\mathrm{THH}(R))^{h(S^1 \times S^1)}.$$

Proof. We apply the categorified Chern character [12, Corollary 6.11] to Perf_R, regarded over finite spectra. It is a symmetric monoidal finitary localizing invariant with values in circle-equivariant objects of $\mathrm{Ind}(S^1 \otimes_{\mathrm{Sp}^\omega} \mathrm{Perf}_R)$. By the cyclic bar construction, we have

$$\mathrm{Ind}(S^1 \otimes_{\mathrm{Sp}^\omega} \mathrm{Perf}_R) \simeq \mathrm{Mod}_{\mathrm{THH}(R)}.$$

By the universal property of $\mathrm{Mot}_R^{\mathrm{loc}}$, we obtain a colimit-preserving symmetric monoidal functor

$$\mathrm{Mot}_R^{\mathrm{loc}} \longrightarrow (\mathrm{Mod}_{\mathrm{THH}(R)})^{hS^1}.$$

The circle acts on $\mathrm{Mod}_{\mathrm{THH}(R)}$ semilinearly for loop rotation on $\mathrm{THH}(R)$. The underlying functor is strongly continuous by [17, Observation 4.51], since $\mathrm{Mot}_R^{\mathrm{loc}}$ is rigid [7, Theorem 3.1(ii)] and the unit of $\mathrm{Mod}_{\mathrm{THH}(R)}$ is compact. Applying continuous K-theory, we obtain a map $K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}}) \to K(\mathrm{THH}(R))$.

The circle-equivariant structure of this functor is a coherent family of equivalences of strongly continuous functors. Continuous K-theory carries it to an equivariant structure on this map, and we obtain a map

$$K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}}) \longrightarrow K(\mathrm{THH}(R))^{hS^1}.$$

The circle-invariant Dennis trace $K(B) \to \mathrm{THH}(B)^{hS^1}$ is natural in the ring spectrum B [12, Remark 6.12]. Hence, for $B = \mathrm{THH}(R)$, it is also equivariant for the first circle. Taking homotopy fixed points for that action, we obtain the required map. Both transformations are multiplicative. □

3.2 The detecting rings

Definition 3.2 (Tate construction of the connective cover)**.** For a spectrum X, set

$$t_+X = (\tau_{\geq 0}X)^{tC_p}$$

with trivial C_p-action on $\tau_{\geq 0}X$. Write $t_+^0X = X$ and $t_+^{k+1}X = t_+(t_+^kX)$, each time with a new trivial C_p-action commuting with the earlier residual actions.

Restricting from the trivial circle action, we obtain a residual S^1/C_p-action on t_+X, and an action of $(S^1/C_p)^k$ on t_+^kX [21, Remark 5.6]. If X is a commutative ring spectrum, then so is t_+X, and the map $\tau_{\geq 0}X \to t_+X$ is a ring map [21, Section 2.1].

Proposition 3.3 (Two Tate constructions)**.** *If E is a height-m Lubin–Tate theory with $m \geq 2$, then*

$$L_{T(m)}K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_{t_+^2E}) \neq 0, \qquad L_{T(j)}t_+^2E = 0 \quad (j > m-2). \tag{3.1}$$

We follow Yuan's argument, using the trace of Proposition 3.1 in place of the iterated Dennis trace on $K(K(-))$. For the first claim, we detect height m in $(t_+^2E)^{h(S^1/C_p)^2}$ as in [21, Lemma 5.7]. For the second, we use [21, Lemmas 5.3 and 5.5] to reduce the bound on coefficients to rank-two blueshift [2, Theorem 3.5].

Proof. Since t_+^2E carries an action of $(S^1/C_p)^2$ by ring maps, we can extend the identity of t_+^2E to a ring map $\mathrm{THH}(\mathrm{THH}(t_+^2E)) \to t_+^2E$ that is equivariant for the two circles. For this we use the universal property of THH as the colimit over S^1 of a constant diagram in commutative rings [21, Recollection 3.3], first with an action of the first circle and then with an action of the second. Composing with the trace of Proposition 3.1, we obtain unital maps

$$K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_{t_+^2E}) \longrightarrow \mathrm{THH}(\mathrm{THH}(t_+^2E))^{h(S^1)^2} \longrightarrow (t_+^2E)^{h(S^1/C_p)^2}. \tag{3.2}$$

By Lemma 2.3, the first claim follows if the target is not $T(m)$-acyclic. Following [21, Lemma 5.7], we show by induction on $k \geq 0$ that $(t_+^k E)^{h(S^1/C_p)^k}$ is not $T(m)$-acyclic. For $k = 0$, this is $L_{T(m)}E \neq 0$.

Let $k \geq 1$ and put $Y = \tau_{\geq 0} t_+^{k-1} E$, with the action of the first $k-1$ residual circles and a new trivial circle action. The idea of the inductive step is that Y is a retract of Y^{tS^1}, and that the Tate orbit lemma identifies Y^{tS^1} with $(t_+^k E)^{h(S^1/C_p)}$ after p-completion.

The retraction. The coefficient maps from $\tau_{\geq 0}E$ to the successive Tate constructions are induced by constant cochains followed by the map from homotopy fixed points to Tate constructions. Hence they are equivariant for the residual circles, where $\tau_{\geq 0}E$ carries the trivial action. It follows that the complex orientation of $\tau_{\geq 0}E$ induces an orientation of Y compatible with the first $k-1$ circles, whose Euler class t has degree -2. On each skeleton $\mathbf{CP}^N$, the powers $1, t, \dots, t^N$ form an equivariant basis over Y, compatibly in N. Consequently,

$$Y^{hS^1} \simeq \prod_{i \geq 0} \Sigma^{-2i} Y, \qquad Y^{tS^1} \simeq \operatorname*{colim}_{r \geq 0} \Sigma^{2r} Y^{hS^1}$$

as module spectra with $(S^1/C_p)^{k-1}$-action [21, proof of Proposition 2.6]. Since multiplication by t shifts the coordinates of the product, the projections to the coefficient of t^r on the rth stage are compatible with the transition maps, and they retract the inclusion of the constant coefficient. Together they define an $(S^1/C_p)^{k-1}$-equivariant retraction $Y \to Y^{tS^1} \to Y$ of module spectra.

The Tate orbit lemma. Since Y is connective, the map

$$Y^{tS^1} \longrightarrow (Y^{tC_p})^{h(S^1/C_p)} = (t_+^k E)^{h(S^1/C_p)}$$

is an equivalence after p-completion [21, Proposition 2.5], and it is equivariant for the first $k-1$ circles. Derived p-completion $X \mapsto \varprojlim_r X/p^r$ commutes with homotopy fixed points, since each $X \mapsto X/p^r$ preserves limits. Taking the remaining homotopy fixed points, we find that $(Y^{h(S^1/C_p)^{k-1}})^\wedge_p$ is a retract of $((t_+^k E)^{h(S^1/C_p)^k})^\wedge_p$.

Conclusion. The cofiber of $Y \to t_+^{k-1}E$ is bounded above. Since coconnective spectra are closed under limits, its homotopy fixed points for $(S^1/C_p)^{k-1}$ are bounded above as well, and hence they are $T(m)$-acyclic. By induction, we therefore have $L_{T(m)} Y^{h(S^1/C_p)^{k-1}} \neq 0$. Since p-completion maps are $T(m)$-equivalences for $m \geq 1$, the retraction above shows that $L_{T(m)}(t_+^k E)^{h(S^1/C_p)^k} \neq 0$. The case $k = 2$ is the first claim.

For the coefficient bound, we regard E with trivial action as a Borel genuine C_p^2-spectrum, and we let $e_{C_p^2}$ be its connective cover in the homotopy t-structure. As in [21, proof of Lemma 5.3], there is a ring map $\Phi^{C_p^2} e_{C_p^2} \to t_+^2 E$. By [21, Lemma 5.5], the comparison $\Phi^{C_p^2} e_{C_p^2} \to \Phi^{C_p^2} E$ is a $T(j)$-equivalence for every $j \geq 1$. By rank-two blueshift [2, Theorem 3.5], $L_{T(j)} \Phi^{C_p^2} E = 0$ for $j > m-2$. Every such j satisfies $j \geq 1$, so $L_{T(j)} \Phi^{C_p^2} e_{C_p^2} \simeq L_{T(j)} \Phi^{C_p^2} E = 0$. The ring map above induces a unital map from this zero ring to $L_{T(j)} t_+^2 E$, and therefore $L_{T(j)} t_+^2 E = 0$. □

3.3 Rational coefficients

Corollary 3.4. *If R is p-local and $T(1) \vee T(2)$-acyclic, then*

$$L_{T(2)} K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}}) \xrightarrow{\sim} L_{T(2)} K^{\mathrm{cont}}(\mathrm{Mot}_{R_{\mathbb{Q}}}^{\mathrm{loc}}).$$

Proof. The map $R \to R_{\mathbb{Q}}$ is a $T(0)$-equivalence. It is also a $T(1) \vee T(2)$-equivalence, since both rings are $T(1) \vee T(2)$-acyclic. The assertion follows from Corollary 2.7 with $j = 2$. □

Proposition 3.5. *At every prime p, the spectra $K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_{H\mathbb{Q}})$ and $K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_{H\mathbb{Q}[u^{\pm 1}]})$, with $|u| = 2$, have telescopic height two.*

Proof. For a height-two Lubin–Tate theory E_2, Proposition 3.3 shows that $t_+^2 E_2$ is $T(1)\vee T(2)$-acyclic and that

$$L_{T(2)} K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_{t_+^2 E_2}) \neq 0.$$

Since $t_+^2 E_2$ is p-local, Corollary 3.4 shows that $L_{T(2)} K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_{(t_+^2 E_2)_{\mathbb{Q}}}) \neq 0$. It remains to construct a unital map from $H\mathbb{Q}[u^{\pm 1}]$ to $(t_+^2 E_2)_{\mathbb{Q}}$.

For this we produce a unit of degree two in $t_+^2 E_2$. Composing the complex orientation of E_2 with the map to the Tate construction, we obtain a composite

$$MU \longrightarrow \tau_{\geq 0} E_2 \longrightarrow \tau_{\geq 0}(t_+ E_2),$$

which is a complex orientation of $Y = \tau_{\geq 0}(t_+ E_2)$. Giving MU and Y the trivial C_p-action and using that the Tate construction is lax monoidal [21, Section 2.1], we obtain a unital ring homomorphism

$$\pi_*(MU^{tC_p}) \longrightarrow \pi_*(Y^{tC_p}).$$

Since MU_* is torsion-free and the p-series has linear term pt, multiplication by the p-series on $MU_*[[t]]$ is injective. It follows that the Euler class $t \in \pi_{-2}(MU^{tC_p})$ is invertible [21, Remark 2.2]. Hence its image in $\pi_{-2}(Y^{tC_p}) = \pi_{-2}(t_+^2 E_2)$ is invertible, and its inverse is a unit of degree two in $(t_+^2 E_2)_{\mathbb{Q}}$. Since free commutative $H\mathbb{Q}$-algebras are free graded commutative, this unit defines unital coefficient maps

$$H\mathbb{Q} \longrightarrow H\mathbb{Q}[u^{\pm 1}] \longrightarrow (t_+^2 E_2)_{\mathbb{Q}}.$$

By Lemma 2.3, the $T(2)$-local nonvanishing therefore holds for $H\mathbb{Q}$ and $H\mathbb{Q}[u^{\pm 1}]$. The upper bounds follow from Proposition 2.4. □

Using finite étale invariance, we extend the computation for $\mathbb{Q}$ to the characteristic-zero fields produced by the chromatic Nullstellensatz.

Proposition 3.6. *Let B be a nonzero ordinary commutative $\mathbb{Q}$-algebra. For every $n \geq 0$,*

$$L_{T(n)} K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_{HB}) = 0 \quad \Longleftrightarrow \quad L_{T(n)} K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_{H\mathbb{Q}}) = 0.$$

The same equivalence holds after adjoining and inverting a degree-two generator on both coefficient rings. Consequently, every characteristic-zero field k satisfies

$$\mathrm{ht}_p(K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_{Hk})) = \mathrm{ht}_p(K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_{Hk[u^{\pm 1}]})) = 2.$$

Proof. The implication from $\mathbb{Q}$ to B follows from Lemma 2.3. Conversely, write B as the filtered union of its finitely generated $\mathbb{Q}$-subalgebras B_i. By Lemmas 2.11 and 2.12, we have $L_{T(n)} K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_{HB_i}) = 0$ for some i. A residue field k_i of B_i at a maximal ideal is finite over $\mathbb{Q}$, and Lemma 2.3 applied to $B_i \to k_i$ gives $L_{T(n)} K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_{Hk_i}) = 0$. Since $H\mathbb{Q} \to Hk_i$ is finite faithfully étale, Theorem 2.10 shows that $L_{T(n)} K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_{H\mathbb{Q}}) = 0$.

The same argument applies in the periodic case, since adjoining $u^{\pm 1}$ commutes with filtered colimits and $Hk_i[u^{\pm 1}]/H\mathbb{Q}[u^{\pm 1}]$ is a base change of $Hk_i/H\mathbb{Q}$, hence finite faithfully étale. The assertion about fields follows from Proposition 3.5. □

Theorem 3.7. *If $\mathrm{ht}_p(R) = 0$, then*

$$\mathrm{ht}_p(K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_R)) = 2.$$

Proof. Since $\mathrm{ht}_p(R) = 0$, the rationalization $R_{\mathbb{Q}} = L_{T(0)}R$ is nonzero. By the chromatic Nullstellensatz at height zero [3, Corollary 5.2], there is a unital composite

$$R \longrightarrow R_{\mathbb{Q}} \longrightarrow Hk[u^{\pm 1}]$$

for an algebraically closed characteristic-zero field k. By Proposition 3.6 and Lemma 2.3, we obtain $L_{T(2)}K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}}) \neq 0$. The upper bound follows from Proposition 2.4. □

Remark 3.8. For connective R, a nonperiodic field suffices: use $R \to R_{\mathbb{Q}} \to H\pi_0(R_{\mathbb{Q}}) \to Hk$, where k is the residue field of a maximal ideal of the nonzero $\mathbb{Q}$-algebra $\pi_0(R_{\mathbb{Q}})$. The periodic target also covers nonconnective coefficients.

4 Positive-height Lubin–Tate theories

By the chromatic Nullstellensatz [3, Corollary 5.2], every commutative ring spectrum of height $n \geq 1$ maps to the Lubin–Tate theory $E_n(k, \Gamma_0)$ of a height-n formal group Γ_0 over a perfect field k. Hence the lower bound at height $n \geq 1$ reduces to the following theorem. Throughout this section, k is a perfect field of characteristic p. When the formal group plays no role, we write $E_m(k)$ for the Lubin–Tate theory of some height-m formal group over k.

Theorem 4.1. *Let $n \geq 1$, let k be a perfect field of characteristic p, and let Γ_0 be a height-n formal group over k. Then*

$$L_{T(n+2)}K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_{E_n(k,\Gamma_0)}) \neq 0.$$

We start from $t_+^2 E_{n+2}(k)$, invert a periodicity class, and $K(n)$-localize. By purity, these coefficient maps preserve the nonvanishing of Proposition 3.3 (Lemmas 4.2 and 4.3). The resulting ring has a residue formal group of exact height n. In Lemma 4.8, we adjoin an isomorphism of this formal group with Γ_0 by finite faithfully étale extensions, which preserve the nonvanishing. The Lubin–Tate universal property then gives a map from $E_n(k, \Gamma_0)$.

4.1 A $K(n)$-local detector

Lemma 4.2. *Let $n \geq 1$. Choose a height-$(n+2)$ Lubin–Tate theory $E_{n+2}(k)$, and let $u \in \pi_2 E_{n+2}(k)$ be a periodicity class. Its image in $t_+^2 E_{n+2}(k)$ makes $(t_+^2 E_{n+2}(k))[u^{-1}]$ an L_n-local $E_{n+2}(k)$-algebra satisfying*

$$\begin{aligned} &L_{T(n+2)}K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_{(t_+^2 E_{n+2}(k))[u^{-1}]}) \\ &\quad \simeq L_{T(n+2)}K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_{t_+^2 E_{n+2}(k)}) \neq 0. \end{aligned}$$

Proof. The ring map $\tau_{\geq 0}E_{n+2}(k) \to t_+E_{n+2}(k)$ factors through $\tau_{\geq 0}t_+E_{n+2}(k)$, which maps to $t_+^2 E_{n+2}(k)$. Hence $t_+^2 E_{n+2}(k)$ is a $\tau_{\geq 0}E_{n+2}(k)$-algebra. Since $E_{n+2}(k) \simeq (\tau_{\geq 0}E_{n+2}(k))[u^{-1}]$, we obtain

$$(t_+^2 E_{n+2}(k))[u^{-1}] \simeq (t_+^2 E_{n+2}(k)) \otimes_{\tau_{\geq 0}E_{n+2}(k)} E_{n+2}(k).$$

The cofiber of $\tau_{\geq 0}E_{n+2}(k) \to E_{n+2}(k)$ is bounded above, hence $T(j)$-acyclic for $j \geq 1$. Since $L_{T(j)}$ is symmetric monoidal and preserves the bar construction, base change along this $T(j)$-equivalence induces an equivalence

$$L_{T(j)}(t_+^2 E_{n+2}(k)) \xrightarrow{\simeq} L_{T(j)}((t_+^2 E_{n+2}(k))[u^{-1}]) \qquad (j \geq 1). \tag{4.1}$$

Hence Corollary 2.7 with $j = n+2$ gives the asserted equivalence, and Proposition 3.3 with $m = n+2$ gives the nonvanishing.

It remains to show that $(t_+^2 E_{n+2}(k))[u^{-1}]$ is L_n-local. Since L_{n+2} is smashing, every $E_{n+2}(k)$-module is L_{n+2}-local. By (4.1) and Proposition 3.3, the ring $(t_+^2 E_{n+2}(k))[u^{-1}]$ is $K(n+1)$- and $K(n+2)$-acyclic. Applying the chromatic fracture square twice, we conclude that it is L_n-local. □

Lemma 4.3 (Morava localization of coefficients)**.** *Let $1 \le n < m$, and let X be an L_n-local commutative $E_m(k)$-algebra. The coefficient map $X \to L_{K(n)}X$ induces a natural equivalence*

$$L_{T(n+2)}K^{\mathrm{cont}}(\mathrm{Mot}_X^{\mathrm{loc}}) \xrightarrow{\simeq} L_{T(n+2)}K^{\mathrm{cont}}(\mathrm{Mot}_{L_{K(n)}X}^{\mathrm{loc}}). \tag{4.2}$$

Proof. Via the complex orientation of $E_m(k)$, the p-local ring X is an $MU_{(p)}$-module, hence a retract of $MU_{(p)} \otimes X$. Since $MU_{(p)}$ splits into suspensions of BP, the spectrum $MU_{(p)} \otimes X$ is a BP-module. By Hovey's comparison of $T(n)$- and $K(n)$-localization for BP-modules [11, Theorem 1.9 and Corollary 1.10], which passes to retracts, we have $L_{T(n)}X \simeq L_{K(n)}X$; see also [21, Section 4.1]. Thus $X \to L_{K(n)}X$ is a $T(n)$-equivalence of p-local rings. Both rings are $T(n+1)$- and $T(n+2)$-acyclic, since they are L_n-local. Hence the assertion follows from Corollary 2.7 with $j = n+2$. □

Combining Lemmas 4.2 and 4.3 (with $m = n+2$), we see that the $K(n)$-local $E_{n+2}(k)$-algebra

$$C = L_{K(n)}((t_+^2 E_{n+2}(k))[u^{-1}])$$

satisfies

$$L_{T(n+2)}K^{\mathrm{cont}}(\mathrm{Mot}_C^{\mathrm{loc}}) \neq 0.$$

In particular, $C \neq 0$.

4.2 Prescribed Lubin–Tate realization

Definition 4.4 (Quillen formal group)**.** Let A be a two-periodic complex orientable commutative ring spectrum. Its *Quillen formal group* is the one-dimensional commutative formal group

$$\operatorname{Spf} \pi_0(A^{\mathbf{CP}_+^\infty}) \quad \text{over } \pi_0 A,$$

with group law induced by tensor product of complex line bundles. Here Spf uses the topology from the finite skeleta of $\mathbf{CP}^\infty$. Its *geometric residue formal groups* are the base changes along ring maps from $\pi_0 A$ to algebraically closed fields.

For a height-m Lubin–Tate theory $E_m(k)$, we choose height parameters, that is, a coordinate on its Quillen formal group and elements $a_1, \dots, a_{m-1}$ as in the setup for [2, Lemma 3.7], with

$$\pi_0 E_m(k) = W(k)[[a_1, \dots, a_{m-1}]], \qquad I_t = (p, a_1, \dots, a_{t-1}) \quad (1 \le t \le m),$$

such that the following holds. Write $a_0 = p$, $a_m = 1$ and $I_0 = 0$. For $0 \le t \le m$, the p-series satisfies $[p](x) \equiv \varepsilon_t a_t x^{p^t}$ modulo I_t and x^{p^t+1} for a unit ε_t. We also set $I_{m+1} = (1)$. For a coefficient algebra D, we write D/I_t for $D/(I_t D)$.

Definition 4.5 (Exact height)**.** A one-dimensional commutative formal group with coordinate over an $\mathbf{F}_p$-algebra D has *exact height* $t \ge 1$ if its p-series is congruent to $c\,x^{p^t}$ modulo x^{p^t+1} with c a unit of D. This condition is independent of the coordinate. A one-dimensional commutative formal group over a characteristic-zero field has height zero.

For $1 \le t \le m$ and an $E_m(k)$-algebra B, the Quillen formal group of B has exact height t over $\pi_0 B/I_t$ if and only if a_t is a unit in $\pi_0 B/I_t$.

By the Lubin–Tate universal property [21, Theorem 4.2], a ring map $E_n(k,\Gamma_0) \to C$ amounts to a map $k \to \pi_0 C/I_n$ together with an isomorphism from the base change of Γ_0 to the residue Quillen formal group of C. Here the map comes from $E_{n+2}(k)$. To obtain the isomorphism, we adjoin an ind-finite étale extension of the residue ring and lift it to C in Lemma 4.8. This uses the exact height and the henselian property proved in the next two lemmas.

Lemma 4.6. *Let B be a nonzero $\mathbb{E}_\infty$-$E_m(k)$-algebra and let $0 \le t \le m$. Then B is L_t-local if and only if $I_{t+1}\pi_0 B = (1)$, equivalently if all geometric residue formal groups have height at most t.*

Proof. By [2, Lemma 3.7(i)–(ii)], applied to the affine stack $\operatorname{Spec} B$ over $\operatorname{Spec} E_m(k)$, B is L_t-local if and only if $\operatorname{Spec}\pi_0 B$ avoids $V(I_{t+1})$, that is, $I_{t+1}\pi_0 B = (1)$. The equivalent condition on geometric residue formal groups follows from the height formula [2, Equation (3.8)]. □

Lemma 4.7. *Let $1 \le n < m$, and let C be a nonzero $K(n)$-local $\mathbb{E}_\infty$-$E_m(k)$-algebra. Then $\pi_0 C$ is derived I_n-complete, and $(\pi_0 C, I_n\pi_0 C)$ is a henselian pair. Moreover, $\pi_0 C/I_n \neq 0$, and the induced formal group over $\pi_0 C/I_n$ has exact height n.*

Proof. Let $x = a_i$ with $0 \le i < n$ be a generator of I_n. Since $x \in I_{i+1}$, we see from Lemma 4.6 that $E_m(k)[1/x]$ is L_i-local, hence $K(n)$-acyclic. The spectrum

$$\operatorname{Hom}_{E_m(k)}(E_m(k)[1/x], C) \simeq \operatorname{holim}(C \xleftarrow{x} C \xleftarrow{x} \cdots)$$

is therefore $K(n)$-local and an $E_m(k)[1/x]$-module, hence zero. From the Milnor sequences, we obtain

$$\varprojlim(\pi_0 C \xleftarrow{x} \pi_0 C \xleftarrow{x} \cdots) = 0, \qquad \varprojlim{}^1(\pi_0 C \xleftarrow{x} \pi_0 C \xleftarrow{x} \cdots) = 0,$$

that is, the derived limit $T(\pi_0 C, x) = R\varprojlim(\cdots \xrightarrow{x} \pi_0 C \xrightarrow{x} \pi_0 C)$ vanishes. The elements $x \in \pi_0 C$ with $T(\pi_0 C, x) = 0$ form a radical ideal [20, Tag 091Q]. It contains the generators of $I_n\pi_0 C$, hence all of it, so $\pi_0 C$ is derived I_n-complete [20, Section 091N]. Consequently, the pair $(\pi_0 C, I_n\pi_0 C)$ is henselian [20, Tag 0G3H].

Since the pair is henselian, we have $I_n\pi_0 C \subseteq \operatorname{Jac}(\pi_0 C)$. As $\pi_0 C \neq 0$, it follows that $\pi_0 C/I_n \neq 0$. Because C is L_n-local, Lemma 4.6 with $t = n$ shows that $I_n\pi_0 C + (a_n) = \pi_0 C$. As $I_n\pi_0 C$ lies in the Jacobson radical, we conclude that a_n is a unit. Hence the Quillen formal group over $\pi_0 C/I_n$ has exact height n. □

Lemma 4.8 (Prescribed Lubin–Tate realization)**.** *Let $1 \le n < m$, let C be a nonzero $K(n)$-local $\mathbb{E}_\infty$-$E_m(k)$-algebra, and let Γ_0 be a height-n formal group over k. There is a filtered diagram of finite faithfully étale C-algebras C_r, with $C_0 = C$, and a unital map*

$$E_n(k,\Gamma_0) \longrightarrow L_{K(n)}C_\infty, \qquad C_\infty = \operatorname*{colim}_r C_r.$$

Both C_∞ and $L_{K(n)}C_\infty$ are nonzero L_n-local $E_m(k)$-algebras.

We construct the extension modulo I_n by Lazard's theorem and lift it to C by henselianity. By faithful flatness, the colimit stays nonzero after $K(n)$-localization.

Proof. By Lemma 4.7, $(\pi_0 C, I_n \pi_0 C)$ is henselian and the Quillen formal group Γ_C over $\pi_0 C/I_n$ has exact height n. The map $W(k) \to \pi_0 C$ induces a map $k \to \pi_0 C/I_n$. Choose coordinates on Γ_C and $\Gamma_0 \otimes_k (\pi_0 C/I_n)$. By Lazard's theorem [16, Theorem 1], the algebra classifying isomorphisms of these two laws of exact height n is a filtered colimit of injective finite étale maps

$$\pi_0 C/I_n = \overline{D}_0 \longrightarrow \overline{D}_1 \longrightarrow \overline{D}_2 \longrightarrow \cdots .$$

Each of these maps is faithfully flat, since it is finite étale and injective.

We now lift this tower to C. Since the pair $(\pi_0 C, I_n \pi_0 C)$ is henselian, reduction modulo I_n induces an equivalence

$$\mathrm{FinEt}(\pi_0 C) \xrightarrow{\simeq} \mathrm{FinEt}(\pi_0 C/I_n)$$

of categories of finite étale algebras [20, Tag 09ZL]. Hence the tower lifts to a tower of finite étale $\pi_0 C$-algebras D_r with $D_0 = \pi_0 C$ and $D_r/I_n \simeq \overline{D}_r$. Each D_r is faithful over $\pi_0 C$, since its rank is positive at every maximal ideal, and every maximal ideal contains $I_n \pi_0 C$. By [15, Theorem 7.5.0.6], this diagram lifts to finite faithfully étale C-algebras C_r with $\pi_0 C_r \simeq D_r$.

The ring $\mathrm{colim}_r D_r$ is faithfully flat over $\pi_0 C$, since its fibers over residue fields are filtered colimits of nonzero rings. For every C-module M, we have

$$\pi_*(M \otimes_C C_\infty) \simeq \pi_* M \otimes_{\pi_0 C} \operatorname*{colim}_r D_r.$$

We take $M = K(n) \otimes C$, which is nonzero because C is nonzero and $K(n)$-local. By the formula above and faithful flatness, we obtain $K(n) \otimes C_\infty \neq 0$, and hence $L_{K(n)} C_\infty \neq 0$.

Since $\pi_0 C_\infty/I_n \simeq \mathrm{colim}_r \overline{D}_r$, this quotient carries the map from k and an isomorphism from the base change of Γ_0 to its Quillen formal group. We transport both along the map

$$\pi_0 C_\infty/I_n \longrightarrow \pi_0(L_{K(n)} C_\infty)/I_n.$$

Since $L_{K(n)} C_\infty$ is a two-periodic complex oriented $K(n)$-local ring, the Lubin–Tate universal property [21, Theorem 4.2] shows that the transported data define a map $E_n(k, \Gamma_0) \to L_{K(n)} C_\infty$. Finally, C_∞ is L_n-local as a module over C, and $L_{K(n)} C_\infty$ is L_n-local as a $K(n)$-local spectrum. □

Proof of Theorem 4.1. We apply Lemma 4.8 to the ring C above and to Γ_0. Since the $T(n+2)$-localization of $K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_C)$ is nonzero, Corollary 2.13 shows that

$$L_{T(n+2)} K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_{C_\infty}) \neq 0.$$

By (4.2) with $m = n+2$, the same holds with C_∞ replaced by $L_{K(n)} C_\infty$. Applying Lemma 2.3 to the unital map

$$K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_{E_n(k,\Gamma_0)}) \longrightarrow K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_{L_{K(n)} C_\infty})$$

induced by the realization map, we obtain the theorem. □

4.3 Proof of double redshift

Proof of Theorem 1.3. Put $n = \mathrm{ht}_p(R)$. The case $n = 0$ is Theorem 3.7. Suppose that $n \geq 1$. Then the commutative ring $L_{T(n)} R_{(p)}$ is nonzero, and the chromatic Nullstellensatz [3, Corollary 5.2] gives a unital composite

$$R \longrightarrow R_{(p)} \longrightarrow L_{T(n)} R_{(p)} \longrightarrow E_n(k, \Gamma_0)$$

for an algebraically closed field k of characteristic p. Theorem 4.1 shows that the spectrum

$$L_{T(n+2)} K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_{E_n(k,\Gamma_0)})$$

is nonzero. By Lemma 2.3, we conclude that $L_{T(n+2)} K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_R) \neq 0$. The upper bound follows from Proposition 2.4. □

5 Continuous indexed pairings

The proof of finite étale invariance (Theorem 2.10) needs a map from the norm of the K-theory of a category to the K-theory of its indexed tensor product. Hilman and Ramzi construct it for compactly generated genuine categories. Since categories of motives are only dualizable, we extend the pairing to continuous K-theory of dualizable genuine categories (Theorem 5.6).

Recollections 5.1 (Genuine categories)**.** Let G be a finite group. A *genuine G-category* is a functor from the opposite of the orbit category of G to categories. Its value at G/H is the *H-fiber* $\mathcal{C}^H$, and its functoriality consists of restrictions $\mathrm{res}^K_H : \mathcal{C}^K \to \mathcal{C}^H$ and conjugations. A G-functor is a natural transformation. The left and right adjoints of res^K_H, when they exist, are the indexed coproduct Ind^K_H and indexed product Coind^K_H.

A genuine G-category is *G-presentable* if its fibers are presentable and it admits genuine colimits, that is, fiberwise colimits preserved by restriction and indexed coproducts satisfying base change along double cosets. A pointed genuine G-category with finite indexed products and coproducts is *G-semiadditive* if the canonical maps from indexed coproducts to indexed products are equivalences. It is *G-stable* if it is G-semiadditive and its fibers are stable [10, Recollections 2.1.5 and 2.1.9, Definition 2.1.10]. We write $\mathrm{Pr}^L_{G,\mathrm{st}}$ for the category of genuine G-presentable G-stable categories and genuine colimit-preserving functors.

The genuine G-category of spectra $\underline{\mathrm{Sp}}$ has H-fiber Sp^H and indexed coproducts the additive inductions Ind^K_H.

By [10, Theorem B.0.1], evaluation at the top fiber is an equivalence

$$\mathrm{Pr}^L_{G,\mathrm{st}} \simeq \mathrm{Mod}_{\mathrm{Sp}^G}(\mathrm{Pr}^L_{\mathrm{st}}).$$

We use this module model below.

Conventions 5.2 (Dualizable modules)*.* For a rigid presentable stable symmetric monoidal category $\mathcal{E}$, let $\mathrm{Cat}^{\mathrm{dual}}_{\mathcal{E}}$ denote the category of dualizable objects of $\mathrm{Mod}_{\mathcal{E}}(\mathrm{Pr}^L_{\mathrm{st}})$ and strongly continuous $\mathcal{E}$-linear functors. Since $\mathcal{E}$ is rigid, an $\mathcal{E}$-linear colimit-preserving functor is strongly continuous if and only if its right adjoint preserves colimits, and that right adjoint is then $\mathcal{E}$-linear [6, Definition 1.10 and Proposition 1.12]. We write $\mathrm{Cat}^{\mathrm{dual}}_G = \mathrm{Cat}^{\mathrm{dual}}_{\mathrm{Sp}^G}$ and regard its objects as genuine G-categories.

A *short exact sequence* in $\mathrm{Cat}^{\mathrm{dual}}_G$ is a fully faithful strongly continuous functor followed by its presentable Verdier quotient, with all terms and maps in $\mathrm{Cat}^{\mathrm{dual}}_G$.

Our inductive arguments use isotropy separation, which detects objects by their restrictions to proper subgroups and their image in a geometric quotient.

Construction 5.3 (Categorical isotropy separation)**.** Let K be a finite group, $\underline{\mathcal{C}}$ a genuine K-presentable stable category with top fiber $\mathcal{C}^K$, and $\mathcal{F}$ a family of subgroups of K, closed under subgroups and conjugation. Let

$$\mathcal{L}_{\mathcal{F}} = \mathrm{Loc}\{\mathrm{Ind}^K_J X \mid J \in \mathcal{F},\ X \in \mathcal{C}^J\} \subseteq \mathcal{C}^K$$

be the localizing subcategory generated by transfers from members of $\mathcal{F}$, and let

$$\Phi^{\mathcal{F}} : \mathcal{C}^K \longrightarrow \Phi^{\mathcal{F}}\mathcal{C} = \mathcal{C}^K/\mathcal{L}_{\mathcal{F}}$$

be the presentable Verdier quotient. Precomposition with $\Phi^{\mathcal{F}}$ identifies colimit-preserving functors out of $\Phi^{\mathcal{F}}\mathcal{C}$ with the colimit-preserving functors out of $\mathcal{C}^K$ that vanish on every $\mathrm{Ind}^K_J X$ with $J \in \mathcal{F}$. The quotient has a fully faithful right adjoint $i_{\mathcal{F}}$.

For the family $\mathcal{P}_K$ of proper subgroups, write

$$q_K = \Phi^{\mathcal{P}_K} : \mathcal{C}^K \longrightarrow \mathcal{C}^{K,\mathrm{geom}} = \Phi^{\mathcal{P}_K}\mathcal{C}$$

and call q_K the *geometric quotient*.

Let $E\mathcal{F}$ be the universal K-space for $\mathcal{F}$, with $(E\mathcal{F})^J \simeq *$ for $J \in \mathcal{F}$ and $(E\mathcal{F})^J = \emptyset$ otherwise, and

$$\widetilde{E}\mathcal{F} = \mathrm{cofib}(\Sigma^\infty_+ E\mathcal{F} \longrightarrow \mathbb{S}_K) \in \mathrm{Sp}^K.$$

Using the Sp^K-action on $\mathcal{C}^K$, we have

$$i_{\mathcal{F}}\Phi^{\mathcal{F}} X \simeq \widetilde{E}\mathcal{F} \otimes X \qquad (X \in \mathcal{C}^K)$$

[10, Construction 2.2.6], [9, Construction 2.2.13]. Indeed, the object $\Sigma^\infty_+ E\mathcal{F} \otimes X$ is built from the cells $\Sigma^\infty_+ K/J \otimes X \simeq \mathrm{Ind}^K_J \mathrm{res}^K_J X$ with $J \in \mathcal{F}$, and therefore lies in $\mathcal{L}_{\mathcal{F}}$. Conversely, if $J \in \mathcal{F}$, then $\mathrm{res}^K_J \widetilde{E}\mathcal{F} = 0$, and therefore $\widetilde{E}\mathcal{F} \otimes \mathrm{Ind}^K_J Y \simeq \mathrm{Ind}^K_J(\mathrm{res}^K_J \widetilde{E}\mathcal{F} \otimes Y) = 0$. It follows that the cofiber sequence

$$\Sigma^\infty_+ E\mathcal{F} \otimes X \to X \to \widetilde{E}\mathcal{F} \otimes X$$

is the localization sequence for $\mathcal{L}_{\mathcal{F}}$.

For $\mathcal{F} = \mathcal{P}_K$, write $i_K = i_{\mathcal{P}_K}$. By adjunction, the essential image of i_K is

$$\mathcal{C}^{K,\mathrm{geom}} \simeq \{Y \in \mathcal{C}^K \mid \mathrm{res}^K_J Y = 0 \text{ for every } J < K\}.$$

For $\underline{\mathcal{C}} = \underline{\mathrm{Sp}}$, we recover $q_K = \Phi^K : \mathrm{Sp}^K \to \mathrm{Sp}$, the smashing localization with idempotent algebra $\widetilde{E}\mathcal{P}_K$. In general, $i_K q_K \simeq \widetilde{E}\mathcal{P}_K \otimes (-)$ by the formula above, and hence q_K is extension of scalars along geometric fixed points:

$$\mathcal{C}^{K,\mathrm{geom}} \simeq \mathrm{Sp} \otimes_{\mathrm{Sp}^K, \Phi^K} \mathcal{C}^K.$$

Lemma 5.4 (Isotropy separation)**.** *In the situation of Construction 5.3, an object $X \in \mathcal{C}^K$ is zero if and only if $\mathrm{res}^K_J X = 0$ for every $J < K$ and $q_K X = 0$. Consequently, a map in $\mathcal{C}^K$ is an equivalence if and only if its restrictions to all proper subgroups and its image under q_K are equivalences.*

Proof. Suppose that $\mathrm{res}^K_J X = 0$ for every $J < K$ and that $q_K X = 0$. By the localization sequence of Construction 5.3, we have $X \simeq \Sigma^\infty_+ E\mathcal{P}_K \otimes X$. This object is built from the cells $\mathrm{Ind}^K_J \mathrm{res}^K_J X$ with $J < K$, which vanish by assumption [10, Observation 2.2.7]. Hence $X = 0$. For maps, apply this to the cofiber, using that restrictions and q_K are exact. □

Suppose $\mathcal{C}^K$ is presentably symmetric monoidal and its transfers satisfy the projection formula $\mathrm{Ind}^K_J Y \otimes X \simeq \mathrm{Ind}^K_J(Y \otimes \mathrm{res}^K_J X)$. Then $\mathcal{L}_{\mathcal{F}}$ is a tensor ideal, and hence q_K is symmetric monoidal. For relative motives, the projection formula is (6.3). We prove the compatibility of the geometric quotient with norms in Lemma 6.12 and Proposition 6.15.

Definition 5.5 (Presentable indexed tensor)**.** For $H \leq G$, let $f : G/H \to G/G$ be the orbit map. The genuine indexed product f_* is right adjoint to restriction from genuine G-categories to genuine H-categories. For a genuine H-category $\mathcal{C}$, its K-fiber is

$$(f_*\mathcal{C})^K \simeq \prod_{[g] \in H\backslash G/K} \mathcal{C}^{H \cap gKg^{-1}}.$$

We recall *f-distributivity* [10, Definition 2.1.13]. Let $f : W \to V$ be a map of finite G-sets, let $\mathcal{C}$ be a W-category and $\mathcal{D}$ a V-category, both admitting genuine colimits, and let $F : f_*\mathcal{C} \to \mathcal{D}$ be a V-functor. For every map of finite G-sets $u : U \to V$, form the pullback

$$\begin{array}{ccc} W' = U \times_V W & \xrightarrow{u'} & W \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ U & \xrightarrow{u} & V. \end{array}$$

For every small genuine W'-category $\mathcal{A}$ and every W'-colimit cocone $d : \mathcal{A}^{\triangleright} \to u'^*\mathcal{C}$, require the cocone

$$(f'_*\mathcal{A})^{\triangleright} \xrightarrow{\mathrm{can}} f'_*(\mathcal{A}^{\triangleright}) \xrightarrow{f'_*d} f'_*(u'^*\mathcal{C}) \simeq u^* f_*\mathcal{C} \xrightarrow{u^*F} u^*\mathcal{D}$$

to be a genuine U-colimit cocone.

For the orbit map $f : G/H \to G/G$ and a genuine H-presentable stable category $\mathcal{C}$, the *presentable indexed tensor* $\mathrm{N}_H^{G,\mathrm{Pr}}\mathcal{C}$ carries a universal f-distributive map

$$\mu_f : f_*\mathcal{C} \longrightarrow \mathrm{N}_H^{G,\mathrm{Pr}}\mathcal{C}.$$

For every genuine G-presentable stable category $\mathcal{D}$, precomposition with μ_f gives an equivalence

$$\mathrm{Fun}_G^L(\mathrm{N}_H^{G,\mathrm{Pr}}\mathcal{C}, \mathcal{D}) \simeq \mathrm{Fun}_G^{f\text{-distributive}}(f_*\mathcal{C}, \mathcal{D}).$$

The functor $f_\otimes$ of Hilman and Ramzi has this universal property [10, Fact D.0.2(2)]. For example, the norm of spectra $\mathrm{N}_H^G : \mathrm{Sp}^H \to \mathrm{Sp}^G$ is f-distributive. Since $\mathrm{N}_H^{G,\mathrm{Pr}} : \mathrm{Pr}^L_{H,\mathrm{st}} \to \mathrm{Pr}^L_{G,\mathrm{st}}$ is a symmetric monoidal 2-functor [10, Proposition D.0.4], it restricts to a functor $\mathrm{Cat}_H^{\mathrm{dual}} \to \mathrm{Cat}_G^{\mathrm{dual}}$. We construct $\mathrm{N}_H^{G,\mathrm{Pr}}$ in the module model in Appendix A.

Let K_H denote the genuine nonconnective K-theory of compact objects, defined on genuine compactly generated H-categories. Its G-lax symmetric monoidal structure comes from norms on equivariant localizing motives [10, Theorems 3.2.1–3.2.2, Corollary 3.3.3]. Since Sp^H is rigid, Efimov's linear extension theorem shows that K_H extends uniquely to a localizing invariant

$$K_H^{\mathrm{cont}} : \mathrm{Cat}_H^{\mathrm{dual}} \longrightarrow \mathrm{Sp}^H$$

[6, Theorem 1.19]. This is the Sp^H-linear version of Definition 2.2.

Theorem 5.6 (Continuous indexed pairings)**.** *For every $H \leq G$, with orbit map $f : G/H \to G/G$, there is a natural pairing*

$$\mathrm{N}_H^G K_H^{\mathrm{cont}}(\mathcal{C}) \longrightarrow K_G^{\mathrm{cont}}(\mathrm{N}_H^{G,\mathrm{Pr}}\mathcal{C}), \qquad \mathcal{C} \in \mathrm{Cat}_H^{\mathrm{dual}}. \tag{5.1}$$

It is natural for strongly continuous functors and agrees with the Hilman–Ramzi pairing on compactly generated categories. If $\mathcal{C}$ is a symmetric monoidal Sp^H-module with compact unit, it takes the norm of the unit to the unit.

To prove Theorem 5.6, we resolve $\mathcal{C}$ by compactly generated categories. A Calkin resolution $\mathcal{C} \to \mathcal{D} \to \mathcal{Q}$ with $\mathcal{D}$ and $\mathcal{Q}$ compactly generated expresses $K_H^{\mathrm{cont}}(\mathcal{C})$ as the fiber of $K_H(\mathcal{D}) \to K_H(\mathcal{Q})$. The norm of a fiber is the total fiber of an indexed cube of smash products of $K_H(\mathcal{D})$ and $K_H(\mathcal{Q})$. The pairings for compactly generated categories map this cube to the K-theory of the corresponding cube of indexed tensor products of $\mathcal{D}$ and $\mathcal{Q}$. It remains to prove excision, namely that the total fiber of the latter cube is $K_G^{\mathrm{cont}}(\mathrm{N}_H^{G,\mathrm{Pr}}\mathcal{C})$ (Proposition 5.11). We prove this by induction on proper subgroups, using isotropy separation (Lemma 5.4). On geometric fixed points, the indexed cube becomes an ordinary cube.

5.1 A uniform Calkin resolution

Every dualizable category is ω_1-compactly generated [5, Corollary 1.21]. Moreover, strongly continuous functors preserve ω_1-compact objects, since their right adjoints preserve colimits. We therefore use the single cardinal ω_1 throughout, and the resolution below is functorial for all strongly continuous functors.

Construction 5.7 (Continuous Calkin resolution)**.** Let $\mathcal{C} \in \mathrm{Cat}_H^{\mathrm{dual}}$. Write $\mathcal{C}^{\omega_1}$ for the genuine H-category of ω_1-compact objects. The colimit functor $\mathrm{Ind}(\mathcal{C}^{\omega_1}) \to \mathcal{C}$ has a fully faithful strongly continuous left adjoint [6, Proposition 1.17].

We use this left adjoint to regard $\mathcal{C}$ as a subcategory of $\mathrm{Ind}(\mathcal{C}^{\omega_1})$, and we define the *continuous Calkin category* by

$$\mathrm{Calk}_{\omega_1}^{\mathrm{cont}}\,\mathcal{C} = \big(\mathrm{Ind}(\mathcal{C}^{\omega_1})/\mathcal{C}\big)^{\omega}.$$

The quotient $\mathrm{Ind}(\mathcal{C}^{\omega_1})/\mathcal{C}$ is compactly generated. Hence we obtain a short exact sequence [6, Proposition 1.17]

$$0 \longrightarrow \mathcal{C} \longrightarrow \mathrm{Ind}(\mathcal{C}^{\omega_1}) \longrightarrow \mathrm{Ind}\big(\mathrm{Calk}_{\omega_1}^{\mathrm{cont}}\,\mathcal{C}\big) \longrightarrow 0, \tag{5.2}$$

which is functorial on $\mathrm{Cat}_H^{\mathrm{dual}}$. Its last map is induced by an exact functor on compact objects.

Since K_H^{cont} is localizing and extends K_H, the sequence (5.2) shows that $K_H^{\mathrm{cont}}(\mathcal{C})$ is the fiber of $K_H(\mathrm{Ind}(\mathcal{C}^{\omega_1})) \to K_H(\mathrm{Ind}(\mathrm{Calk}_{\omega_1}^{\mathrm{cont}}\,\mathcal{C}))$.

We next compare continuous K-theory with restriction, induction and fixed points. The functors res_H^G, additive induction Ind_H^G and fiber evaluation $\mathcal{C} \mapsto \mathcal{C}^H$ preserve these short exact sequences and compact generation. The linear extension theorem also extends natural transformations [6, Theorem 1.19]. Hence the comparisons for compactly generated K-theory extend to continuous K-theory, together with their compatibility with composition and conjugation. In particular, we have

$$\begin{aligned} \mathrm{res}_H^G\, K_G^{\mathrm{cont}}(\mathcal{C}) &\simeq K_H^{\mathrm{cont}}(\mathrm{res}_H^G\,\mathcal{C}), \\ K_G^{\mathrm{cont}}(\mathrm{Ind}_H^G\,\mathcal{C}) &\simeq \mathrm{Ind}_H^G\, K_H^{\mathrm{cont}}(\mathcal{C}), \\ \big(K_G^{\mathrm{cont}}(\mathcal{C})\big)^H &\simeq K^{\mathrm{cont}}(\mathcal{C}^H), \end{aligned} \tag{5.3}$$

with categorical fixed points on the left. Let $H < G$. Since Φ^G vanishes on spectra induced from proper subgroups, the induction comparison shows that $\Phi^G K_G^{\mathrm{cont}}(\mathrm{Ind}_H^G\,\mathcal{C}) = 0$.

5.2 Excision for norms

A norm does not preserve fibers in general. Instead, the norm of a fiber is computed by a cubical formula, and the excision proposition below is the categorical counterpart of this formula for continuous K-theory.

Definition 5.8 (Genuine cubes)**.** Let I be a finite G-set. We regard its subset poset $\mathcal{P}(I)$, with G acting by $g \cdot J = gJ$, as a genuine G-category with fiber

$$\mathcal{P}(I)^K = \{J \subseteq I \mid kJ = J \text{ for every } k \in K\}$$

at G/K, ordered by inclusion [10, Recollections 2.1.18].

For a genuine G-category $\mathcal{D}$, an *I-cube* is a G-functor $X : \mathcal{P}(I) \to \mathcal{D}$. For $J \subseteq I$ write G_J for its stabilizer and $X_J \in \mathcal{D}^{G_J}$ for its value; the value at J in $\mathcal{D}^K$, for $K \leq G_J$, is $\operatorname{res}^{G_J}_K X_J$. If $\mathcal{D}$ is G-presentable and G-stable, define the *total fiber* and *total cofiber* by

$$\operatorname{tfib}_I X = \operatorname{fib}\Big(X_\varnothing \longrightarrow \overset{G}{\lim}_{\varnothing \neq J \subseteq I} X_J\Big),$$
$$\operatorname{tcofib}_I X = \operatorname{cofib}\Big(\overset{G}{\operatorname{colim}}_{J \subsetneq I} X_J \longrightarrow X_I\Big),$$

objects of $\mathcal{D}^G$, where the limit and colimit are genuine indexed ones over the indicated G-posets; compare [4, Definition 3.3].

Lemma 5.9 (Total fibers and total cofibers)**.** *For every I-cube X in a genuine G-presentable G-stable category, there is a natural equivalence*

$$\operatorname{tfib}_I X \simeq S^{-\mathbb{R}[I]} \otimes \operatorname{tcofib}_I X. \tag{5.4}$$

Here $\mathbb{R}[I]$ is the permutation representation with basis I. The equivalence is compatible with restriction to subgroups.

Proof. For $A \in \mathcal{D}^G$, let $\delta_\varnothing A$ be the cube with value A at $\varnothing$ and zero elsewhere. A map $\delta_\varnothing A \to X$ amounts to a map $A \to X_\varnothing$ together with a coherent nullhomotopy on the nonempty vertices. Hence $\delta_\varnothing$ is left adjoint to tfib_I. The total cofiber of $\delta_\varnothing A$ is $S^{\mathbb{R}[I]} \otimes A$, since the cubical model has one suspension coordinate for each element of I, permuted by G, and $[0,1]^I/\partial[0,1]^I \simeq S^{\mathbb{R}[I]}$ [4, Examples 1.17–1.19]. Applying tcofib_I to the counit, we obtain a natural comparison map

$$S^{\mathbb{R}[I]} \otimes \operatorname{tfib}_I X \simeq \operatorname{tcofib}_I(\delta_\varnothing \operatorname{tfib}_I X) \longrightarrow \operatorname{tcofib}_I X,$$

and we show that it is an equivalence by filtering the cube.

For $-1 \leq r \leq |I|$, let $X^{\leq r}$ be the right Kan extension of X restricted to the vertices of cardinality at most r; it has value X_J for $|J| \leq r$ and zero otherwise. The projections form a tower from $X = X^{\leq |I|}$ to $X^{\leq -1} = 0$. The fiber of $X^{\leq r} \to X^{\leq r-1}$ is supported on the vertices of cardinality r and splits as a finite sum over their G-orbits. Since total fiber and total cofiber are exact, it suffices to check the comparison on a cube supported on one orbit.

Choose a vertex J in that orbit, put $H = G_J$, and let $Y \in \mathcal{D}^H$ be its value. Since colimits over the orbit are additive inductions from H and limits over it are coinductions, we may compute with the H-cube supported at J. In this cube the J-directions are terminally supported and the directions in $I \setminus J$ are initially supported. Computing the endpoint contributions and applying Fubini, we obtain

$$\operatorname{tfib}_I X \simeq \operatorname{Coind}^G_H(S^{-\mathbb{R}[J]} \otimes Y), \qquad \operatorname{tcofib}_I X \simeq \operatorname{Ind}^G_H(S^{\mathbb{R}[I \setminus J]} \otimes Y).$$

By genuine stability, Ind^G_H is equivalent to Coind^G_H. Using the projection formula and the decomposition $\operatorname{res}^G_H \mathbb{R}[I] = \mathbb{R}[J] \oplus \mathbb{R}[I \setminus J]$, we see that the comparison map is the identity in the complementary directions and the evaluation equivalence $S^{\mathbb{R}[J]} \otimes S^{-\mathbb{R}[J]} \otimes Y \to Y$ in the J-directions. Hence the comparison map is an equivalence on each layer, and therefore on X. Finally, its construction uses only indexed limits, indexed colimits and the action of spheres, all of which commute with restriction to subgroups. □

For trivial G, the equivalence (5.4) reads $\operatorname{tfib}_I X \simeq \Omega^{|I|} \operatorname{tcofib}_I X$. We shall use two consequences of Lemma 5.9.

(a) *Norms of fibers.* Let $x \to y$ be a map in a stable H-category, and let N be a distributive functor out of the indexed product over $f : G/H \to G/G$, such as the norm N_H^G. Write $N(x \to y)$ for the G/H-cube with value N applied to y in the positions of J and to x outside J. Then $N(\mathrm{cofib}(x \to y)) \simeq \mathrm{tcofib}_{G/H} N(x \to y)$ [10, Lemma 2.1.19]. For the spectrum norm and a map $x \to y$ in Sp^H, we have

$$\mathrm{N}_H^G \,\mathrm{fib}(x \to y) \simeq S^{-\mathbb{R}[G/H]} \otimes \mathrm{N}_H^G \,\mathrm{cofib}(x \to y),$$

because N_H^G is symmetric monoidal, $\mathrm{fib} = \Sigma^{-1} \,\mathrm{cofib}$, and $\mathrm{N}_H^G(S^{-1}) \simeq S^{-\mathbb{R}[G/H]}$. Combining this with Lemma 5.9, we obtain

$$\mathrm{N}_H^G \,\mathrm{fib}(x \to y) \simeq \mathrm{tfib}_{G/H} \,\mathrm{N}_H^G(x \to y).$$

(b) *Geometric fixed points.* Let X be a cube of genuine G-spectra. The functor Φ^G is exact and symmetric monoidal, with $\Phi^G S^{-\mathbb{R}[I]} \simeq S^{-|I/G|}$. On total cofibers, Φ^G sees only the G-invariant subsets, because the cells of the colimit indexed by chains with proper stabilizer are properly induced [10, proof of Proposition 3.2.3]. The G-invariant subsets are exactly the unions of orbits. Combining Lemma 5.9 with the ordinary comparison of fibers and cofibers, we therefore identify $\Phi^G \,\mathrm{tfib}_I X$ with the total fiber of the $|I/G|$-cube $J \mapsto \Phi^G X_J$, where J runs over the unions of orbits.

Construction 5.10 (Quotient cubes)**.** Write $I = \coprod_r G/H_r$ and choose a short exact sequence $\mathcal{C}_r \hookrightarrow \mathcal{D}_r \to \mathcal{Q}_r$ in $\mathrm{Cat}_{H_r}^{\mathrm{dual}}$ for each orbit.

The *quotient cube* V is the following G-functor from $\mathcal{P}(I)$ to the genuine category with fibers $\mathrm{Cat}_K^{\mathrm{dual}}$. For $J \in \mathcal{P}(I)^K$, decompose I into K-orbits. To an orbit represented by gH_r we attach the conjugate of the rth sequence, restricted to $K \cap gH_rg^{-1}$. We apply the indexed tensor from this stabilizer to K to the quotient input if the orbit lies in J and to the middle input otherwise, and we take the tensor product of these factors over Sp^K. For $J \subseteq J'$ the arrow uses the quotient maps on the orbits in $J' \setminus J$ and the identity elsewhere. The restriction, conjugation and composition maps are those of the universal distributive functor for indexed tensor. The endpoint values in the G-fiber are

$$V_\emptyset = \bigotimes_r \mathrm{N}_{H_r}^{G,\mathrm{Pr}} \mathcal{D}_r, \qquad V_I = \bigotimes_r \mathrm{N}_{H_r}^{G,\mathrm{Pr}} \mathcal{Q}_r.$$

Applying K_K^{cont} in every subgroup fiber, we obtain the cube $K_G^{\mathrm{cont}}(V)$ of genuine spectra.

The inclusions $\mathcal{C}_r \to \mathcal{D}_r$ induce an *augmentation*

$$\bigotimes_r \mathrm{N}_{H_r}^{G,\mathrm{Pr}} \mathcal{C}_r \longrightarrow V_\emptyset.$$

We obtain its coherent nullhomotopies by applying the universal indexed tensor operations to the [2]-diagrams $\mathcal{C}_r \to \mathcal{D}_r \to \mathcal{Q}_r$, together with the factorizations of their composites through the zero category. At a nonempty vertex, at least one position uses such a composite. Hence the tensor functor factors through an indexed tensor product with a zero factor, and this product is zero because distributivity includes preservation of the empty colimit. Applying K_K^{cont} in every subgroup fiber to these data, we obtain a map of I-cubes

$$\delta_\varnothing K_G^{\mathrm{cont}} \left(\bigotimes_r \mathrm{N}_{H_r}^{G,\mathrm{Pr}} \mathcal{C}_r \right) \longrightarrow K_G^{\mathrm{cont}}(V).$$

Let α be its adjoint under $\delta_\varnothing \dashv \mathrm{tfib}_I$ from Lemma 5.9.

Proposition 5.11 (Excision for norms)**.** *Let $I = \coprod_r G/H_r$ be a nonempty finite G-set and let V be the $\mathcal{P}(I)$-indexed genuine quotient cube defined above from short exact sequences $\mathcal{C}_r \hookrightarrow \mathcal{D}_r \to \mathcal{Q}_r$ in $\mathrm{Cat}^{\mathrm{dual}}_{H_r}$. The augmentation induces an equivalence*

$$\alpha : K^{\mathrm{cont}}_G\left(\bigotimes_r \mathrm{N}^{G,\mathrm{Pr}}_{H_r}\mathcal{C}_r\right) \xrightarrow{\simeq} \mathrm{tfib}_I\, K^{\mathrm{cont}}_G(V). \tag{5.5}$$

The same assertion holds after tensoring the cube and its augmentation with a fixed dualizable genuine G-category.

For G trivial and I a single point, Proposition 5.11 is the localization sequence $K^{\mathrm{cont}}(\mathcal{C}) \to K^{\mathrm{cont}}(\mathcal{D}) \to K^{\mathrm{cont}}(\mathcal{Q})$. For $G = C_2$ and $I = C_2/e$, a single sequence $\mathcal{C} \hookrightarrow \mathcal{D} \to \mathcal{Q}$ gives, in the e-fiber, the square

$$\begin{array}{ccc} \mathcal{D}\otimes\mathcal{D} & \longrightarrow & \mathcal{Q}\otimes\mathcal{D} \\ \downarrow & & \downarrow \\ \mathcal{D}\otimes\mathcal{Q} & \longrightarrow & \mathcal{Q}\otimes\mathcal{Q}. \end{array}$$

The group C_2 exchanges the two middle vertices, and the C_2-fiber of $\mathcal{P}(I)$ contains only $\emptyset$ and I, with values $\mathrm{N}^{C_2,\mathrm{Pr}}_e\mathcal{D}$ and $\mathrm{N}^{C_2,\mathrm{Pr}}_e\mathcal{Q}$. The total fiber of $K^{\mathrm{cont}}_{C_2}(V)$ also involves the middle vertices through the e-fiber, and the equivalence (5.5) identifies it with $K^{\mathrm{cont}}_{C_2}(\mathrm{N}^{C_2,\mathrm{Pr}}_e\mathcal{C})$.

We prove Proposition 5.11 at the end of this subsection by induction on $|G|$ for proper subgroups. On geometric fixed points, consequence (b) reduces the claim to an ordinary cube. What we need there is that $\Phi^G K^{\mathrm{cont}}_G$ of an indexed tensor product is exact in each variable (Lemma 5.18). To prove this exactness, we need the colimits in the indexed tensor cube to be formed along fully faithful inclusions. Lemma 5.12 gives a criterion, and Proposition 5.15 applies it to the indexed tensor cube and computes the layers of the resulting filtration.

Lemma 5.12 (Colimits of fully faithful inclusions)**.** *Let $\mathcal{E}$ be a rigid presentable stable symmetric monoidal category, let P be a finite nonempty poset with binary meets, and let $s \mapsto \mathcal{C}_s$ be a diagram in $\mathrm{Cat}^{\mathrm{dual}}_{\mathcal{E}}$. Suppose it has a cocone $i_s : \mathcal{C}_s \to \mathcal{B}$ of fully faithful strongly continuous functors, with $\mathcal{B}$ dualizable. Write $a_{r,s}$ for the transition functor when $r \le s$. Assume that for every $s, t \in P$ and $r = s \wedge t$, the canonical Beck–Chevalley mate*

$$a_{r,t}a^R_{r,s} \longrightarrow i^R_t i_s \tag{5.6}$$

is an equivalence. Then

$$\operatorname*{colim}_{s\in P}{}^{\mathrm{Pr}^L_{\mathcal{E},\mathrm{st}}}\mathcal{C}_s \longrightarrow \mathcal{B}$$

is fully faithful and strongly continuous. This colimit is dualizable and is also the colimit in $\mathrm{Cat}^{\mathrm{dual}}_{\mathcal{E}}$.

Proof. The inclusion $\mathrm{Cat}^{\mathrm{dual}} \to \mathrm{Pr}^L_{\mathrm{st}}$ preserves colimits [5, Propositions 1.7 and 1.65]. The relative assertion follows from $\mathrm{Cat}^{\mathrm{dual}}_{\mathcal{E}} \simeq \mathrm{Mod}_{\mathcal{E}}(\mathrm{Cat}^{\mathrm{dual}})$ [6, Proposition 1.12(v)], because colimits of modules are created on underlying objects. Hence the colimit $\mathcal{A}$, its structure maps $j_s : \mathcal{C}_s \to \mathcal{A}$, and the induced map $L : \mathcal{A} \to \mathcal{B}$ all belong to $\mathrm{Cat}^{\mathrm{dual}}_{\mathcal{E}}$.

It remains to prove that L is fully faithful. Recall that colimits in $\mathrm{Pr}^L_{\mathrm{st}}$ are computed as limits along the right adjoints. By (5.6), the transformation $\mathcal{C}_{s\wedge(-)} \to \mathcal{C}_{(-)}$ of diagrams on P is right adjointable. Since $t \mapsto s \wedge t$ is final onto $P_{\le s}$, which has terminal object s, the colimit of the source diagram is $\mathcal{C}_s$. Passing to colimits, we obtain

$$j^R_t j_s \simeq a_{s\wedge t,t}\, a^R_{s\wedge t,s} \simeq i^R_t i_s.$$

This is the argument of [10, proof of Lemma 3.2.5], which is stated for posets with binary joins but uses only the meets $s \wedge t$. Thus the mate $j_t^R \to i_t^R L$ is an equivalence on the image of every j_s. Since both functors preserve colimits, it is an equivalence on all of $\mathcal{A}$. We now apply the jointly conservative family $(j_t^R)_t$ to the unit $\mathrm{id} \to L^R L$ and find that the unit is an equivalence on the image of each j_s. As L^R preserves colimits, the unit is an equivalence everywhere, and L is fully faithful. □

The tensor filtrations needed here and in Section 6 are instances of a single construction, the relative indexed tensor, which requires the following structure on the coefficients.

Definition 5.13 (Indexed distributivity)**.** Let $\mathcal{C}$ be a genuine G-category admitting all genuine colimits and equipped with a genuine symmetric monoidal structure. For $K \leq G$ and a finite family $H_1, \dots, H_r \leq K$, the indexed multiplication associated with $\coprod_a K/H_a \to K/K$ is

$$\prod_a \mathcal{C}^{H_a} \longrightarrow \mathcal{C}^K, \qquad (X_a)_a \longmapsto \bigotimes_a \mathrm{N}_{H_a}^K X_a.$$

The structure is *distributive* if every indexed multiplication is distributive in the sense of Definition 5.5 [10, Definitions 2.1.13 and 2.1.14].

Definition 5.14 (Relative rigidity)**.** Let $\mathcal{V}$ be a presentably symmetric monoidal stable category and let $\mathcal{A}$ be a commutative algebra in $\mathrm{Mod}_{\mathcal{V}}(\mathrm{Pr}^L_{\mathrm{st}})$. The algebra $\mathcal{A}$ is *locally rigid over* $\mathcal{V}$ if it is dualizable over $\mathcal{V}$ and the multiplication $\mathcal{A} \otimes_{\mathcal{V}} \mathcal{A} \to \mathcal{A}$ has a colimit-preserving $\mathcal{A}$-bimodule-linear right adjoint. It is *rigid over* $\mathcal{V}$ if, in addition, the unit functor $\mathcal{V} \to \mathcal{A}$ has a $\mathcal{V}$-linear colimit-preserving right adjoint; in the terminology of [17, Definitions 4.5 and 4.36], its unit is then $\mathcal{V}$-*atomic*. For $\mathcal{V} = \mathrm{Sp}$ this is the notion of rigidity in Conventions 2.1.

Let $\mathcal{E}$ be a genuine G-presentable G-stable category with a distributive genuine symmetric monoidal structure. By Definition 5.5, the indexed multiplication of $\mathcal{E}$ linearizes to commutative algebra maps $\mathrm{N}_H^{K,\mathrm{Pr}}(\mathcal{E}^H) \to \mathcal{E}^K$ for $H \leq K \leq G$, with the base-change and composition coherences of $\mathcal{E}$. If $\mathcal{C}$ is an $\mathcal{E}^H$-module, its indexed tensor over $\mathcal{E}$ is the $\mathcal{E}^K$-module

$$\mathcal{E}^K \otimes_{\mathrm{N}_H^{K,\mathrm{Pr}}(\mathcal{E}^H)} \mathrm{N}_H^{K,\mathrm{Pr}} \mathcal{C}, \tag{5.7}$$

where $\mathrm{N}_H^{K,\mathrm{Pr}}\mathcal{C}$ is a $\mathrm{N}_H^{K,\mathrm{Pr}}(\mathcal{E}^H)$-module by applying the symmetric monoidal 2-functor $\mathrm{N}_H^{K,\mathrm{Pr}}$ to the $\mathcal{E}^H$-action [10, Proposition D.0.4]. The same formula applies to functors and natural transformations.

Assume now that $\mathcal{E}^G$ is rigid over Sp^G. Genuine categories are identified with Sp^G-modules by a symmetric monoidal equivalence [10, Theorem B.0.1]. Under this equivalence, additive induction is restriction of scalars along $\mathrm{Sp}^G \to \mathrm{Sp}^K$, and restriction of genuine categories is its left adjoint. Hence $\mathcal{E}^K \simeq \mathrm{Sp}^K \otimes_{\mathrm{Sp}^G} \mathcal{E}^G$, and it follows that every $\mathcal{E}^K$ is rigid over Sp^K in the sense of Definition 5.14, and therefore over Sp. By [17, Corollary 4.49], the objects and morphisms of $\mathrm{Cat}^{\mathrm{dual}}_{\mathcal{E}^K}$ (Conventions 5.2) are the $\mathcal{E}^K$-module objects and internal left adjoints in $\mathrm{Cat}^{\mathrm{dual}}_K$.

Proposition 5.15 (Tensor-cube filtrations)**.** *Let $\mathcal{E}$ be a genuine G-presentable G-stable category with a distributive genuine symmetric monoidal structure, whose top fiber $\mathcal{E}^G$ is rigid over Sp^G. Let $I = \coprod_r G/H_r$ be a finite nonempty G-set, and choose a short exact sequence*

$$\mathcal{C}_r \longrightarrow \mathcal{D}_r \longrightarrow \mathcal{Q}_r \quad \text{in } \mathrm{Cat}^{\mathrm{dual}}_{\mathcal{E}^{H_r}}$$

for each orbit. Form the genuine tensor cube T indexed by the subset G-poset $\mathcal{P}(I)$, using (5.7): in a direction belonging to G/H_r use $\mathcal{D}_r$ if that direction lies in the subset and $\mathcal{C}_r$

otherwise. More explicitly, at the stabilizer G_J of a subset $J \subseteq I$, restrict and conjugate the input sequences along the G_J-orbits of I, and take their relative indexed tensor products over $\mathcal{E}$. This prescription includes all subgroup fibers and their structure maps.

For every invariant downward-closed family $P \subseteq \mathcal{P}(I)$, the genuine colimit

$$\operatorname*{colim}^{G}_{J \in P} T_J \longrightarrow T_I$$

is fully faithful and strongly continuous. Its source is a dualizable $\mathcal{E}^G$-module. The genuine colimit may be computed in presentable module categories or dualizable module categories, with the same resulting $\mathcal{E}^G$-module; these genuine colimits include additive inductions from all subgroup fibers. For nested invariant downward-closed families $P \subseteq P'$, the induced map between their colimits is also fully faithful and strongly continuous.

In particular, put $F_j = \operatorname{colim}^G_{|J|\leq j} T_J$. For $1 \leq j \leq |I|$ there is a natural equivalence

$$F_j/F_{j-1} \simeq \bigoplus_{\substack{[J]\in G\backslash\mathcal{P}(I)\\ |J|=j}} \operatorname{Ind}^G_{G_J} \left(\left(\bigotimes_{i\in J,\mathcal{E}} \mathcal{Q}_i \right) \otimes_{\mathcal{E}^{G_J}} \left(\bigotimes_{i\notin J,\mathcal{E}} \mathcal{C}_i \right) \right). \tag{5.8}$$

The factors indexed by i in this formula mean the conjugate-restriction families just described, and each tensor is the genuine indexed tensor over $\mathcal{E}$ in the G_J-fiber. Induction is additive induction of these module categories. Thus the quotient of T_I by the full punctured-cube boundary is the indexed tensor with $\mathcal{Q}_r$ in every direction. The filtration and its layer equivalences are natural in the input short exact sequences and commute with subgroup restriction.

Proof. The idea is to prove the proposition first over spectra, where full faithfulness can be checked on proper restrictions and on the geometric quotient, and then to obtain the case of general $\mathcal{E}$ by extension of scalars. Step 1 is the ordinary calculation for the trivial group and $\mathcal{E} = \mathrm{Sp}$. In Step 2, where $\mathcal{E} = \underline{\mathrm{Sp}}$, we prove full faithfulness of the genuine boundary maps by isotropy separation and compute their quotients by distributivity. In Step 3 we show that for general $\mathcal{E}$ all cubes and boundary maps are linear over a common rigid coefficient algebra, and we extend scalars from it.

Step 1: ordinary tensor cubes over spectra. Let $G = e$ and $\mathcal{E} = \mathrm{Sp}$. Taking tensor products of the internal adjunctions of the inputs, we obtain continuous right adjoints to all maps of the cube. For $J, L \subseteq I$, the Beck–Chevalley mate of the intersection square is, in each tensor position, either the identity mate or the invertible unit of $\mathcal{C}_i \hookrightarrow \mathcal{D}_i$. Since nonempty downward-closed families are closed under intersections, Lemma 5.12 over Sp applies and shows that their colimit maps are fully faithful and strongly continuous. The empty family has zero colimit. We now attach the subsets in order of cardinality. Since tensor products preserve colimits in each variable separately, the quotient at J is the tensor product of the $\mathcal{Q}_i$ for $i \in J$ with the $\mathcal{C}_i$ for $i \notin J$. Distinct subsets of the same cardinality meet in the preceding skeleton, and therefore their quotients form a direct sum.

Step 2: genuine boundaries over spectra. Let $\mathcal{E} = \underline{\mathrm{Sp}}$. We prove full faithfulness for every invariant downward-closed P by induction on $|G|$. The normed 2-functor of [10, Proposition D.0.4] preserves duality data, internal adjunctions and invertible units. Hence every vertex is dualizable, and every vertex map to T_I is fully faithful and strongly continuous at its stabilizer. A genuine colimit over a finite poset is built from colimits of modules and additive inductions from stabilizers [10, Proposition C.0.1 and its proof]. Both operations preserve dualizability and strongly continuous functors, so the cocone lies in dualizable categories. As in the proof of Lemma 5.12, the inclusion of dualizable into presentable categories preserves

colimits, and therefore the colimit is also the presentable one. It follows that the boundary map has a continuous internal right adjoint, and it remains to show that the unit of this adjunction is invertible.

We check invertibility of the unit on proper restrictions and on the geometric quotient. Restriction to a proper subgroup $K < G$ splits each G/H_r into K-orbits $K/(K \cap {}^g H_r)$ and yields a tensor cube of the same kind. By induction, the unit is therefore invertible after restriction to every proper subgroup. For the geometric quotient $\Phi^{\mathcal{P}_G}$ of Construction 5.3, with $\mathcal{P}_G$ the family of proper subgroups, we use its description $\mathcal{M} \mapsto \mathrm{Sp} \otimes_{\mathrm{Sp}^G, \Phi^G} \mathcal{M}$ on module categories. This symmetric monoidal 2-functor preserves colimits and internal adjunctions together with their units and counits. By the projection formula, it kills categories additively induced from proper subgroups. We compute the genuine colimit by cells indexed by orbits of chains, where the cell of a chain is induced from its stabilizer [10, proof of Proposition 3.2.3]. Only the chains of G-invariant subsets survive, and hence

$$\Phi^{\mathcal{P}_G}\left(\operatorname*{colim}_{J \in P}{}^{G} T_J\right) \simeq \operatorname*{colim}_{J \in P^G} \Phi^{\mathcal{P}_G} T_J. \tag{5.9}$$

Since invariant subsets are unions of orbits, the right side of (5.9) is a downward-closed part of an ordinary tensor cube over Sp with factor maps $\Phi^{\mathcal{P}_G} \mathrm{N}_{H_r}^{G,\mathrm{Pr}} \mathcal{C}_r \to \Phi^{\mathcal{P}_G} \mathrm{N}_{H_r}^{G,\mathrm{Pr}} \mathcal{D}_r$, which are fully faithful internal left adjoints because the norm and $\Phi^{\mathcal{P}_G}$ preserve internal adjunctions with invertible units. By Step 1, the image of the boundary map under $\Phi^{\mathcal{P}_G}$ is fully faithful. Now let x be an object of the source of the boundary map. The cofiber of the unit at x has zero restriction to every proper subgroup and zero geometric quotient. Applying Lemma 5.4 in the source genuine category, we conclude that this cofiber vanishes. Thus the boundary map is fully faithful.

For the quotients, recall that the universal indexed tensor on genuine presentable stable categories is distributive [8, Theorem 3.65, Corollary 4.7, and Theorem 4.22]. Fix $J \subseteq I$ and work over G_J. The attaching boundary of J is the colimit over the proper subsets of J, with the $\mathcal{C}$-factors outside J held fixed. By full faithfulness, its cofiber is the Verdier quotient. Applying [10, Lemma 2.1.19] to each G_J-orbit of J and using Fubini, we identify this cofiber with the indexed tensor having $\mathcal{Q}$ in J and $\mathcal{C}$ outside J. In the skeletal filtration, the orbit of J contributes the additive induction of this quotient from G_J [10, Lemma 2.1.21], and distinct orbits of the same cardinality meet in the preceding skeleton. This proves (5.8) over spectra.

Step 3: extension of scalars. For general $\mathcal{E}$, we regard the inputs as genuine H_r-presentable stable categories and let T^{abs} be their tensor cube over spectra. From the indexed multiplications of $\mathcal{E}$ we obtain an algebra map

$$\mathcal{A} = \bigotimes_{r, \mathrm{Sp}^G} \mathrm{N}_{H_r}^{G,\mathrm{Pr}}(\mathcal{E}^{H_r}) \longrightarrow \mathcal{E}^G.$$

For $K \leq G$, choose representatives g of $K\backslash G/H_r$ and put $L_{r,g} = K \cap gH_r g^{-1}$. For K-invariant J, let $\mathcal{M}_{r,g}(J)$ be the restriction to $L_{r,g}$ of the conjugate of $\mathcal{D}_r$ if $gH_r \in J$ and of $\mathcal{C}_r$ otherwise. By base change for norms, we have

$$\mathrm{res}_K^G \mathcal{A} \simeq \bigotimes_{r,\,[g],\mathrm{Sp}^K} \mathrm{N}_{L_{r,g}}^{K,\mathrm{Pr}}(\mathcal{E}^{L_{r,g}}), \qquad (T_J^{\mathrm{abs}})^K \simeq \bigotimes_{r,\,[g],\mathrm{Sp}^K} \mathrm{N}_{L_{r,g}}^{K,\mathrm{Pr}} \mathcal{M}_{r,g}(J),$$

with $[g] \in K\backslash G/H_r$. Applying the 2-functors $\mathrm{N}_{L_{r,g}}^{K,\mathrm{Pr}}$ to the module actions, we make each vertex a $\mathrm{res}_K^G \mathcal{A}$-module.

The same construction applies to every chain of subsets over its common stabilizer, since the diagram of such a chain consists of identities and conjugate-restrictions of the linear

functors $\mathcal{C}_r \to \mathcal{D}_r$. The required coherences are provided by [10, Fact D.0.2(2) and Proposition D.0.4]. Using the Beck–Chevalley equivalences for norms and conjugation, we see that these actions are compatible with restriction and with the choice of representatives. Thus T^{abs} is an $\mathcal{A}$-module diagram, including its restriction and conjugation maps.

Colimits of modules are created on underlying categories. For an additive induction, we obtain the action from the projection formula $\mathcal{A} \otimes_{\mathrm{Sp}^G} \mathrm{Ind}_K^G \mathcal{M} \simeq \mathrm{Ind}_K^G(\mathrm{res}_K^G \mathcal{A} \otimes_{\mathrm{Sp}^K} \mathcal{M})$. Hence every boundary colimit, skeletal colimit and presentable quotient of T^{abs} is an $\mathcal{A}$-module, with $\mathcal{A}$-linear structure maps.

The algebra $\mathcal{A}$ is rigid over Sp^G, because $\mathrm{N}_{H_r}^{G,\mathrm{Pr}}$ is a symmetric monoidal 2-functor carrying the rigidity data of $\mathcal{E}^{H_r}$ over Sp^{H_r} to those of $\mathrm{N}_{H_r}^{G,\mathrm{Pr}}(\mathcal{E}^{H_r})$, and finite tensor products preserve them. Likewise $\mathrm{res}_K^G \mathcal{A}$ is rigid over Sp^K. By [17, Proposition 4.17], the vertices and boundaries, which are dualizable over spectra, are dualizable over the corresponding restrictions of $\mathcal{A}$.

Fix an invariant downward-closed P and let $i_P : \mathcal{B}_P = \mathrm{colim}_{J\in P}^G T_J^{\mathrm{abs}} \to T_I^{\mathrm{abs}}$. By Step 2, there is an adjunction $i_P \dashv i_P^R$ in which i_P^R preserves colimits and the unit is invertible. Since i_P is $\mathcal{A}$-linear and $\mathcal{A}$ is locally rigid, it follows from [17, Proposition 4.18] that this is an internal adjunction of $\mathcal{A}$-modules.

Extension of scalars $\mathcal{E}^G \otimes_{\mathcal{A}} (-)$ is a 2-functor preserving colimits and duality data. Hence it carries $i_P \dashv i_P^R$ to an adjunction between colimit-preserving functors with invertible unit, and consequently $\mathcal{E}^G \otimes_{\mathcal{A}} i_P$ is fully faithful and strongly continuous.

It remains to identify $\mathcal{E}^G \otimes_{\mathcal{A}} T^{\mathrm{abs}}$ with T. Restriction commutes with relative tensor products, and for K-invariant J associativity of the relative tensor product gives

$$\mathcal{E}^K \otimes_{\mathrm{res}_K^G \mathcal{A}} (T_J^{\mathrm{abs}})^K \simeq \bigotimes_{r,\,[g],\mathcal{E}^K} \left(\mathcal{E}^K \otimes_{\mathrm{N}_{L_{r,g}}^{K,\mathrm{Pr}}(\mathcal{E}^{L_{r,g}})} \mathrm{N}_{L_{r,g}}^{K,\mathrm{Pr}} \mathcal{M}_{r,g}(J) \right),$$

whose factors are the relative indexed tensors (5.7). This equivalence is natural in the module functors, and therefore it identifies all vertices and face maps. For additive induction, we have $\mathcal{E}^G \otimes_{\mathcal{A}} \mathrm{Ind}_K^G \mathcal{M} \simeq \mathrm{Ind}_K^G(\mathcal{E}^K \otimes_{\mathrm{res}_K^G \mathcal{A}} \mathcal{M})$ by the projection formula applied to the bar construction. Hence extension of scalars carries the genuine boundary colimits, the skeletal colimits and the pushouts defining the quotients to those of T, and it carries the layer formula over spectra to (5.8). The dualizable and presentable genuine colimits agree by [17, Corollary 4.49]. For $P \subseteq P'$, the induced map is a morphism of dualizable modules. It is fully faithful because its composite with the inclusion of the P'-colimit into T_I is the inclusion of the P-colimit. Finally, all constructions commute with the input functors and with restriction to subgroups. □

Corollary 5.16 (Relative tensor filtrations)**.** *Let $\mathcal{E}$ be a rigid presentable stable symmetric monoidal category, and let $\mathcal{C}_i \hookrightarrow \mathcal{D}_i$, for i in a finite nonempty set I, be fully faithful strongly continuous functors of dualizable $\mathcal{E}$-modules. Form the ordinary tensor cube*

$$T_J = \left(\bigotimes_{i\in J,\mathcal{E}} \mathcal{D}_i \right) \otimes_{\mathcal{E}} \left(\bigotimes_{i\notin J,\mathcal{E}} \mathcal{C}_i \right), \qquad J \subseteq I.$$

For every downward-closed family $P \subseteq \mathcal{P}(I)$, the map from $\mathrm{colim}_{J\in P} T_J$ to T_I is fully faithful and strongly continuous. The colimit agrees in $\mathrm{Pr}^L_{\mathcal{E},\mathrm{st}}$ and $\mathrm{Cat}^{\mathrm{dual}}_{\mathcal{E}}$. Set $F_j = \mathrm{colim}_{|J|\le j} T_J$; its skeleta satisfy, for $1 \le j \le |I|$,

$$F_j/F_{j-1} \simeq \bigoplus_{\substack{J\subseteq I\\ |J|=j}} \left(\bigotimes_{i\in J,\mathcal{E}} (\mathcal{D}_i/\mathcal{C}_i) \right) \otimes_{\mathcal{E}} \left(\bigotimes_{i\notin J,\mathcal{E}} \mathcal{C}_i \right). \tag{5.10}$$

The quotient by the punctured-cube boundary is $\bigotimes_{i\in I,\mathcal{E}}(\mathcal{D}_i/\mathcal{C}_i)$.

Proof. Take $G = e$ in Proposition 5.15. Then the relative indexed tensor is the ordinary relative tensor product. The presentable quotients $\mathcal{D}_i/\mathcal{C}_i$ are dualizable and the quotient functors are strongly continuous. Hence the sequences $\mathcal{C}_i \hookrightarrow \mathcal{D}_i \to \mathcal{D}_i/\mathcal{C}_i$ are short exact sequences of the kind required in the proposition. □

Corollary 5.17 (Genuine tensor filtrations)**.** *Let $H \leq G$ and let $\mathcal{C} \hookrightarrow \mathcal{D} \to \mathcal{Q}$ be a short exact sequence in $\mathrm{Cat}_H^{\mathrm{dual}}$. For $I = G/H$, let T be the genuine indexed tensor cube with $\mathcal{D}$ in a subset of its directions and $\mathcal{C}$ elsewhere. Then the skeletal inclusions*

$$F_0 = \mathrm{N}_H^{G,\mathrm{Pr}}\mathcal{C} \longrightarrow \cdots \longrightarrow F_{|I|} = \mathrm{N}_H^{G,\mathrm{Pr}}\mathcal{D}$$

are fully faithful and strongly continuous. The quotient by the full punctured-cube boundary is $\mathrm{N}_H^{G,\mathrm{Pr}}\mathcal{Q}$. For $0 < j < |I|$, the layer F_j/F_{j-1} is a finite sum of categories additively induced from proper subgroup stabilizers. The same assertions about arbitrary invariant downward-closed families and the layer formula hold for any finite G-set with independent short exact sequences on its orbits. These genuine colimits are dualizable and agree with their presentable counterparts.

Proof. Take $\mathcal{E} = \underline{\mathrm{Sp}}$ in Proposition 5.15. Since $\mathrm{N}_H^{K,\mathrm{Pr}}\mathrm{Sp}^H \simeq \mathrm{Sp}^K$, formula (5.7) is the absolute indexed tensor. The layers for $0 < j < |I|$ are properly induced because a nonempty proper subset of the transitive G-set G/H has proper stabilizer. □

Lemma 5.18 (Separate geometric exactness)**.** *For a finite collection of subgroups $H_r \leq G$, let $f_r : G/H_r \to G/G$ be the orbit maps. The functor*

$$(\mathcal{C}_r)_r \longmapsto \Phi^G K_G^{\mathrm{cont}}\left(\bigotimes_r \mathrm{N}_{H_r}^{G,\mathrm{Pr}}\mathcal{C}_r\right)$$

is localizing in each variable. The same holds after tensoring the argument with a fixed dualizable genuine G-category.

Proof. Fix r and a short exact sequence $\mathcal{C}_r \hookrightarrow \mathcal{D}_r \to \mathcal{Q}_r$, and apply Corollary 5.17 with $I = G/H_r$. Let $\mathcal{B}$ be the fixed factor, namely the tensor product of the indexed tensor products of the other variables with the fixed dualizable category. Tensoring with $\mathcal{B}$ preserves the skeletal inclusions and their quotients. By the projection formula $(\mathrm{Ind}_K^G \mathcal{E}) \otimes \mathcal{B} \simeq \mathrm{Ind}_K^G(\mathcal{E} \otimes \mathrm{res}_K^G \mathcal{B})$, the layers F_j/F_{j-1} for $0 < j < |I|$ remain properly induced after tensoring with $\mathcal{B}$. These layers are killed by $\Phi^G K_G^{\mathrm{cont}}$, as shown after (5.3). Since K_G^{cont} is localizing, it follows that the map $\Phi^G K_G^{\mathrm{cont}}(F_0 \otimes \mathcal{B}) \to \Phi^G K_G^{\mathrm{cont}}(F_{|I|-1} \otimes \mathcal{B})$ is an equivalence. The last layer is $F_{|I|}/F_{|I|-1} \simeq \mathrm{N}_{H_r}^{G,\mathrm{Pr}}\mathcal{Q}_r$, and we obtain the cofiber sequence

$$\Phi^G K_G^{\mathrm{cont}}(\mathrm{N}_{H_r}^{G,\mathrm{Pr}}\mathcal{C}_r \otimes \mathcal{B}) \to \Phi^G K_G^{\mathrm{cont}}(\mathrm{N}_{H_r}^{G,\mathrm{Pr}}\mathcal{D}_r \otimes \mathcal{B}) \to \Phi^G K_G^{\mathrm{cont}}(\mathrm{N}_{H_r}^{G,\mathrm{Pr}}\mathcal{Q}_r \otimes \mathcal{B}).$$

For an orbit G/G, the norm is the identity, and we use ordinary exactness in that position. Finally, all the functors involved preserve filtered colimits. □

Proof of Proposition 5.11. We argue by induction on $|G|$, proving the statement with the fixed factor $\mathcal{B}$ at the same time. By Lemma 5.4, a map of genuine G-spectra is an equivalence if its restrictions to all proper subgroups and its image under Φ^G are equivalences. Hence it suffices to show that $\mathrm{res}_K^G \alpha$ for $K < G$ and $\Phi^G \alpha$ are equivalences.

For $G = e$, the claim says that K^{cont} of the tensor product of the kernels is the total fiber of the cube formed by the tensor products of the $\mathcal{D}_r$ and the $\mathcal{Q}_r$. This follows by applying exactness of K^{cont} and of $- \otimes \mathcal{B}$ in one position at a time.

Let $K < G$. Restriction to K takes the indexed tensor product over G/H_r to the tensor product over $[g] \in K\backslash G/H_r$ of indexed tensor products over $K/(K \cap {}^g H_r)$ of conjugates of the rth sequence, and the quotient cube over I to the quotient cube over $\mathrm{res}^G_K I$. Some inputs now appear several times. Since the statement for independent inputs specializes along the diagonal to this case, the induction hypothesis shows that $\mathrm{res}^G_K \alpha$ is an equivalence.

It remains to treat $\Phi^G\alpha$. By consequence (b) above, the target $\Phi^G \operatorname{tfib}_I K^{\mathrm{cont}}_G(V)$ is the total fiber of the ordinary cube, with one direction for each orbit G/H_r, whose vertex at a union of orbits J is $\Phi^G K^{\mathrm{cont}}_G$ of the indexed tensor product with $\mathcal{Q}_r$ for the orbits in J and $\mathcal{D}_r$ for the others. We apply Lemma 5.18 to the sequences $\mathcal{C}_r \to \mathcal{D}_r \to \mathcal{Q}_r$, one orbit at a time, and find that this total fiber is $\Phi^G K^{\mathrm{cont}}_G(\bigotimes_r \mathrm{N}^{G,\mathrm{Pr}}_{H_r} \mathcal{C}_r)$, via the same augmentation that induces $\Phi^G\alpha$. Hence $\Phi^G\alpha$ is an equivalence. □

5.3 Construction of continuous indexed pairings

Proof of Theorem 5.6. Applying continuous K-theory to the Calkin resolution (5.2), we obtain the fiber sequence

$$K^{\mathrm{cont}}_H(\mathcal{C}) \longrightarrow K_H\big(\mathrm{Ind}(\mathcal{C}^{\omega_1})\big) \longrightarrow K_H\Big(\mathrm{Ind}\big(\mathrm{Calk}^{\mathrm{cont}}_{\omega_1} \mathcal{C}\big)\Big)\,.$$

By fact (a) on norms of fibers, $\mathrm{N}^G_H K^{\mathrm{cont}}_H(\mathcal{C})$ is therefore the total fiber of a G/H-cube. Its vertex at J is the norm with the factor K_H of the Calkin category in the positions of J and the factor $K_H(\mathrm{Ind}(\mathcal{C}^{\omega_1}))$ in the positions outside J.

Let V be the quotient cube of (5.2). Since the vertices of V are compactly generated, the natural pairings of [10, Corollary 3.3.3] give a map of G/H-cubes from the cube above to $K^{\mathrm{cont}}_G(V)$, whose vertex at J is $K_{G_J}(V_J)$. Passing to total fibers, we obtain a map

$$\mathrm{N}^G_H K^{\mathrm{cont}}_H(\mathcal{C}) \longrightarrow \operatorname{tfib}_{G/H} K^{\mathrm{cont}}_G(V),$$

and the excision equivalence (5.5) identifies its target with $K^{\mathrm{cont}}_G(\mathrm{N}^{G,\mathrm{Pr}}_H \mathcal{C})$. This is the pairing (5.1). It is natural in $\mathcal{C}$, since the Calkin resolution and α are natural for strongly continuous functors and the space of inverses of α is contractible.

Suppose now that $\mathcal{C}$ is compactly generated. Then the compactly generated pairings give a map of augmented cubes. On the augmentations this map is the Hilman–Ramzi pairing of $\mathcal{C}$, and on total fibers it is the pairing just constructed. By naturality of α, the two pairings agree.

It remains to prove unit compatibility. Since the unit of $\mathcal{C}$ is compact, the symmetric monoidal Sp^H-linear functor $\mathrm{Sp}^H \to \mathcal{C}$ is strongly continuous [6, Proposition 1.12(i)]. The pairing is natural along this functor, and indexed tensor products preserve tensor units and unit maps. Hence it suffices to treat Sp^H, where the pairing is the Hilman–Ramzi pairing, which preserves the unit. We conclude that the pairing is unit compatible for every $H \leq G$. □

6 Norms on relative localizing motives

Finite étale invariance (Theorem 2.10) follows from a unit-compatible multiplicative transfer on the continuous K-theory of relative motives. For this we show that finite étale tensor induction descends to motives (Proposition 6.10), that the resulting indexed multiplication

has a linearization (Proposition 6.15), and that this linearization is strongly continuous (Proposition 6.16). Together with Theorem 5.6, this gives (6.11).

6.1 The genuine symmetric monoidal coefficient system

Definition 6.1 (Faithful finite étale torsors)**.** Let G be a finite group and R a commutative ring spectrum. A *faithful finite étale G-torsor* S/R is a commutative R-algebra with a G-action such that $R \to S$ is finite étale, $- \otimes_R S$ is conservative on R-modules, and the canonical maps

$$R \longrightarrow S^{hG}, \qquad S \otimes_R S \longrightarrow \prod_{g \in G} S, \quad a \otimes b \mapsto (a \cdot g(b))_{g \in G},$$

are equivalences. Disconnected torsors are allowed.

Fix such a torsor S/R. For every $H \leq G$, the algebra S^{hH} is finite étale over R. Indeed, since S is a perfect R-module, we have $S^{hH} \otimes_R S \simeq (S \otimes_R S)^{hH} \simeq \prod_{G/H} S$, and the finite étale condition descends along the faithfully flat map $R \to S$. Hence every map $S^{hK} \to S^{hH}$ with $H \leq K$ is finite étale.

Construction 6.2 (The coefficient G-category)**.** The *coefficient G-category associated with* S/R is the genuine G-category $\underline{\mathrm{Cat}}^{\mathrm{perf}}_{S/R}$ with fibers

$$(\underline{\mathrm{Cat}}^{\mathrm{perf}}_{S/R})^H = \mathrm{Cat}^{\mathrm{perf}}_{S^{hH}}, \qquad \underline{\mathrm{Cat}}^{\mathrm{perf}}_{S/R}\left(\coprod_i G/H_i\right) = \prod_i \mathrm{Cat}^{\mathrm{perf}}_{S^{hH_i}}.$$

For the orbit map $f : G/H \to G/K$ with $H \leq K$, the restriction f^* is extension of scalars along $p : S^{hK} \to S^{hH}$, and for $g \in G$ the conjugation from the H-fiber to the ${}^gH$-fiber is extension of scalars along $g : S^{hH} \to S^{h({}^gH)}$. The coherences come from associativity and base change for extension of scalars. Additive transfers and indexed tensor products are constructed below.

For $f : G/H \to G/K$, restriction of coefficients along p defines $f_* : (\underline{\mathrm{Cat}}^{\mathrm{perf}}_{S/R})^H \to (\underline{\mathrm{Cat}}^{\mathrm{perf}}_{S/R})^K$. For $H = e$ and $K = G$, the restriction f^* is $\mathcal{C} \mapsto \mathcal{C} \otimes_R S$, and f_* regards an S-linear category as an R-linear one. By Lemma 6.4, f_* is the additive transfer, and the descent in that lemma is what we use to construct indexed tensor products. In this setting, Karoubi sequences and localizing invariants are meant in the linear sense of Definition 1.2.

Conventions 6.3 (Semilinear actions). Let T'/T be a G-torsor. Coefficient transport $\mathcal{C}' \mapsto \mathcal{C}' \otimes_{T',g} T'$ defines a G-action on $\mathrm{Mod}_{\mathrm{Mod}_{T'}}(\mathrm{Pr}^L_{\mathrm{st}})$. A *semilinear G-action* is an object of $\left(\mathrm{Mod}_{\mathrm{Mod}_{T'}}(\mathrm{Pr}^L_{\mathrm{st}})\right)^{hG}$, that is, a category $\mathcal{C}'$ with coherent g-semilinear self-equivalences. We use the same construction on the $(\infty, 2)$-category of linear colimit-preserving functors and linear natural transformations, on compactly generated categories with compact-preserving functors, and, by passing to compact objects, on small perfect categories.

Lemma 6.4 (Finite étale coefficient change)**.** *For a finite étale map $f : T \to T'$, extension and restriction of coefficients satisfy*

$$f^* \dashv f_* \dashv f^*. \tag{6.1}$$

These adjunctions restrict to compactly generated linear categories and compact-preserving functors, and hence to small perfect categories. If f is faithful, scalar extension detects equivalences and full faithfulness. Linear presentable categories and compactly generated

linear categories satisfy finite étale descent as symmetric monoidal $(\infty, 2)$*-categories, with linear colimit-preserving functors and linear natural transformations. Semilinear descent for a torsor is compatible with tensor products, filtered colimits, and Karoubi sequences. Dualizability and strong continuity can also be checked on the cover.*

Proof. Every assertion rests on the trace of T' over T: it gives ambidexterity, and for faithful f it splits $T \to T'$, so that scalar extension is comonadic. Since f is finite étale, the trace pairing $T' \otimes_T T' \to T$, $a \otimes b \mapsto \mathrm{tr}(ab)$, is perfect. Hence T' is a Frobenius algebra over T, and $\mathrm{Mod}_{T'}$ is a self-dual Mod_T-module, compatibly with the $\mathrm{Mod}_{T'}$-action. For a T-linear category $\mathcal{D}$, the right adjoint of f_* is the cofree extension $\mathrm{Fun}^L_T(\mathrm{Mod}_{T'}, \mathcal{D})$, and the self-duality identifies it with the free extension $\mathcal{D} \otimes_T \mathrm{Mod}_{T'} = f^*\mathcal{D}$. This proves (6.1), also for compactly generated categories and compact-preserving functors, by the Frobenius argument of [1, Lemma 15.4 and the proof of Lemma 15.5].

Suppose that f is faithful. The trace $\pi_0 T' \to \pi_0 T$ is surjective, since it is surjective over each residue field and its finitely generated cokernel vanishes by Nakayama's lemma. Choose $a \in \pi_0 T'$ with $\mathrm{tr}(a) = 1$. Then the map $\mathrm{tr}(a \cdot -)$ splits $T \to T'$ as a map of T-modules. Hence every T-module M is a retract of $M \otimes_T T'$, and in particular $- \otimes_T T'$ is conservative. By the formula

$$\mathrm{Map}_{\mathcal{C} \otimes_T T'}(x \otimes_T T', y \otimes_T T') \simeq \mathrm{Map}_{\mathcal{C}}(x, y) \otimes_T T',$$

scalar extension detects full faithfulness. It also detects equivalences. Suppose that the base-changed functor is an equivalence, so that the functor itself is fully faithful. Every object y of the target is a retract of the underlying object of $y \otimes_T T'$, which lies in the image of the source. Since the essential image is closed under retracts, it contains y.

We now prove descent. By (6.1), scalar extension preserves limits and colimits. Since it is also conservative, it is comonadic by the Barr–Beck–Lurie theorem. Resolving by the comonad $- \otimes_T T'$, we obtain

$$\mathrm{Mod}_{\mathrm{Mod}_T}(\mathrm{Pr}^L_{\mathrm{st}}) \simeq \lim_{[n] \in \Delta} \mathrm{Mod}_{\mathrm{Mod}_{T' \otimes_T (n+1)}}(\mathrm{Pr}^L_{\mathrm{st}}),$$

where the limit is taken over the Amitsur diagram $[n] \mapsto T'^{\otimes_T (n+1)}$. Suppose that a category becomes compactly generated after scalar extension. By (6.1), restriction carries the compact generators of the scalar extension to compact objects, and these objects generate, because every object is a retract of the restriction of its free extension. The same argument shows that scalar extension detects compact-preserving functors. For fixed $\mathcal{C}$, we apply the limit-preserving functor $\mathrm{Fun}^L_T(\mathcal{C}, -)$ to the Amitsur resolution of $\mathcal{D}$. This gives

$$\mathrm{Fun}^L_T(\mathcal{C}, \mathcal{D}) \simeq \lim_{[n] \in \Delta} \mathrm{Fun}^L_T(\mathcal{C}, \mathcal{D} \otimes_T T'^{\otimes_T (n+1)}),$$

and by adjunction the terms of this limit are categories of $T'^{\otimes_T (n+1)}$-linear functors between scalar extensions. Hence descent holds for linear functors and linear natural transformations as well. For ordinary affine coefficients this is [1, Lemma 15.6].

Finally, let T'/T be a G-torsor. Then $T'^{\otimes_T (n+1)} \simeq \prod_{G^n} T'$, and the Amitsur diagram becomes the cobar construction of the G-action. In other words, the descent equivalence is semilinear descent. Since scalar extension commutes with tensor products, colimits and quotients, semilinear descent is compatible with tensor products, filtered colimits and Karoubi sequences. The descent comparison is a symmetric monoidal equivalence of $(\infty, 2)$-categories. Therefore duals, internal right adjoints, and their units and counits descend, which proves the assertions on dualizability and strong continuity. □

Construction 6.5 (Coherent finite étale norms)**.** For $H \leq K$ and $p : S^{hK} \to S^{hH}$, finite étale tensor induction

$$p_\otimes : \mathrm{Mod}_{\mathrm{Mod}_{S^{hH}}}(\mathrm{Pr}^L_{\mathrm{st}}) \longrightarrow \mathrm{Mod}_{\mathrm{Mod}_{S^{hK}}}(\mathrm{Pr}^L_{\mathrm{st}})$$

is constructed by semilinear descent along the K-torsor S/S^{hK} (Lemma 6.4) from

$$p_\otimes(\mathcal{C}) \otimes_{S^{hK}} S \simeq \bigotimes_{kH \in K/H,\, S} (\mathcal{C} \otimes_{S^{hH},k} S), \tag{6.2}$$

where $k : S^{hH} \to S$ is the canonical map followed by the automorphism k of S. Since $S^{hH} \to S$ is H-invariant, the factor depends only on the coset kH, so the factors are indexed by K/H. For $\ell \in K$, the equivalences

$$(\mathcal{C} \otimes_{S^{hH},k} S) \otimes_{S,\ell} S \simeq \mathcal{C} \otimes_{S^{hH},\ell k} S$$

together with the permutation $kH \mapsto \ell kH$ define a semilinear K-action on the right side, and by Lemma 6.4 the right side descends to a category $p_\otimes(\mathcal{C})$. The same construction applies to dualizable linear presentable categories.

We obtain the composition and base-change equivalences for $p_\otimes$, with all higher coherences, from the sheaf construction of Bachmann and Hoyois. Let FEt_R be the category of affine finite étale R-schemes with the finite étale topology. By [15, Theorem 7.5.0.6], it is equivalent to the corresponding category for $\pi_0 R$. Hence FEt_R is extensive, and finite étale sheaves take finite disjoint unions to products. Moreover, every map in FEt_R is finite étale, and therefore becomes a sum of fold maps after a finite étale cover. These are the hypotheses of [1, Corollary C.13], with m the class of all maps. For $\mathcal{D}$ with finite products, that corollary extends every finite étale sheaf $F : \mathrm{FEt}_R^{\mathrm{op}} \to \mathrm{CAlg}(\mathcal{D})$ uniquely to

$$\widehat{F} : \mathrm{Span}(\mathrm{FEt}_R) \longrightarrow \mathrm{CAlg}(\mathcal{D}),$$

whose covariant functoriality along the fold map $\coprod_1^d X \to X$ is the d-ary multiplication of $F(X)$. For a map $\operatorname{Spec} T' \to \operatorname{Spec} T$, this covariant functoriality is the norm $F(T') \to F(T)$, which is determined locally by the fold maps.

We apply this with $\mathcal{D} = \mathrm{Cat}_{(\infty,2)}$ and $F(T)$ the symmetric monoidal $(\infty,2)$-category of T-linear presentable categories, linear colimit-preserving functors and linear natural transformations. By Lemma 6.4, F is a finite étale sheaf. For a split map $T \to T^d$, the covariant functoriality is the d-fold tensor product over T. Since $S^{hH} \otimes_{S^{hK}} S \simeq \prod_{K/H} S$, the value of $\widehat{F}$ at $\operatorname{Spec} p$ becomes, on this cover, the tensor product of the conjugates. Hence it is the functor $p_\otimes$ of (6.2). Since $p_\otimes$ is a symmetric monoidal $(\infty,2)$-functor, it preserves internal adjunctions and dualizable objects. For connective coefficients this is [1, Lemma 15.6 and the paragraph after it]. For nonconnective coefficients, we use the descent of Lemma 6.4 instead.

The subfunctor of compactly generated categories and compact-preserving functors is again a finite étale sheaf by Lemma 6.4. It is preserved by tensor products on a splitting cover, since tensor products of compact generators are compact generators. We apply [1, Corollary C.13] to this subfunctor and pass to compact objects. This gives coherent norms on small perfect categories.

These norms equip $\underline{\mathrm{Cat}}^{\mathrm{perf}}_{S/R}$ with a genuine G-symmetric monoidal structure. Indeed, the functor $\widehat{F}$ of Construction 6.5, restricted to the algebras S^{hH} and their products, encodes norms along all orbit maps. These norms are compatible with composition, and with restriction by the double-coset formula. We denote the structure norm by

$$\mathrm{N}^K_{H,S/R} \simeq p_\otimes : (\underline{\mathrm{Cat}}^{\mathrm{perf}}_{S/R})^H = \mathrm{Cat}^{\mathrm{perf}}_{S^{hH}} \longrightarrow \mathrm{Cat}^{\mathrm{perf}}_{S^{hK}} = (\underline{\mathrm{Cat}}^{\mathrm{perf}}_{S/R})^K,$$

and use the same notation on dualizable linear presentable categories. By Lemma 6.4, the underlying coefficient system with the additive transfers f_* is genuine semiadditive.

The norms $\mathrm{N}^K_{H,S/R}$ on small perfect categories preserve filtered colimits. By (6.2), it suffices to check this on the cover. There the norm is a tensor product of $|K/H|$ conjugates composed with the diagonal, and the diagonal of a filtered category is cofinal. On presentable coefficient categories, relative tensor products preserve colimits separately in each variable. Hence, on the splitting cover, the indexed norm takes families of colimit cocones to colimit cocones, compatibly with base change. By Lemma 6.4, the same holds after descent. Therefore the norms are distributive in the sense of Definition 5.13.

To apply Proposition 5.15, we also use module spectra as coefficients. The corresponding genuine coefficient category has J-fiber $\mathrm{Mod}_{S^{hJ}}$. Its restrictions are extensions of scalars, and its additive transfers are restrictions of scalars. As in Lemma 6.4, the trace pairing gives ambidexterity, and base change for tensor products gives the double-coset formula. It is genuinely semiadditive, since scalar extension from S^{hK} to S is conservative and, via $S^{hH} \otimes_{S^{hK}} S \simeq \prod_{K/H} S$, carries the comparison $w_! \to w_*$ for an orbit map to the map from a finite sum of S-modules to their product. Since the fibers are presentable and stable, the genuine category is G-presentable and G-stable [10, Recollections 2.1.3, 2.1.5, 2.1.9 and Definition 2.1.10]. Its indexed multiplication is tensor induction. On the cover S, it is given by the tensor product over S of the factors $M \otimes_{S^{hH},k} S$ for $kH \in K/H$, with the semilinear permutation action. Finite étale descent for modules follows from the trace and comonadic argument of Lemma 6.4, and the coherences come from [1, Corollary C.13] applied to module categories.

We check that these coefficients satisfy the hypotheses of Proposition 5.15. The structure is distributive, since on the splitting cover S tensor products of modules preserve colimits separately, and extension to S is conservative and preserves colimits. Moreover, $\mathrm{Mod}_{S^{hJ}}$ is rigid over Sp, and under [10, Theorem B.0.1] it is a commutative Sp^J-algebra. Since Sp^J is also rigid over Sp, it follows from [17, Corollary 4.46] that $\mathrm{Mod}_{S^{hJ}}$ is rigid over Sp^J.

For these coefficients, the relative indexed tensor (5.7) of an S^{hH}-linear category $\mathcal{D}$ is

$$\mathrm{Mod}_{S^{hK}} \otimes_{\mathrm{N}^{K,\mathrm{Pr}}_H(\mathrm{Mod}_{S^{hH}})} \mathrm{N}^{K,\mathrm{Pr}}_H \mathcal{D}.$$

We identify it with $p_\otimes \mathcal{D}$, compatibly with restriction and with maps in each tensor position. Under [10, Theorem B.0.1], genuine restriction from K to $L \leq K$ is scalar extension along $\mathrm{Sp}^K \to \mathrm{Sp}^L$ (Appendix A). Hence the restriction of an S^{hK}-linear category is its extension of coefficients to S^{hL}. In particular, restriction to the trivial subgroup is extension to S. After this restriction, the double-coset formula for $\mathrm{N}^{K,\mathrm{Pr}}_H$ gives

$$\begin{aligned} &\mathrm{Mod}_S \otimes_{\bigotimes_{kH\in K/H} \mathrm{Mod}_S} \left(\bigotimes_{kH\in K/H} (\mathcal{D} \otimes_{S^{hH},k} S) \right) \\ &\quad \simeq \bigotimes_{kH\in K/H,\, S} (\mathcal{D} \otimes_{S^{hH},k} S), \end{aligned}$$

where the tensor products in the first line are over Sp. The equivalence balances the S-action in every position. It is natural in the inputs, in their linear functors and in natural transformations. The second line is the right side of (6.2).

Balancing commutes with permutation of positions and with the coefficient transport $(\mathcal{D} \otimes_{S^{hH},k} S) \otimes_{S,\ell} S \simeq \mathcal{D} \otimes_{S^{hH},\ell k} S$ for $\ell \in K$. Since the coherences on both sides come from the same base-change equivalences, the comparison is semilinearly K-equivariant, and by Lemma 6.4 it descends to an equivalence with $p_\otimes \mathcal{D}$.

Next we check compatibility with restriction to a subgroup $L \leq K$. Decompose K/H into L-orbits indexed by $[g] \in L\backslash K/H$, and put $H_g = L \cap gHg^{-1}$. By [10, Fact D.0.2(2) and Proposition D.0.4], the restriction of the absolute indexed tensor is the tensor product over Sp^L of the indexed tensors from H_g to L, with coefficient input $\mathrm{Mod}_{S^{hH_g}}$ and category input $\mathcal{D} \otimes_{S^{hH},g} S^{hH_g}$. Hence the restricted relative indexed tensor is the tensor product over $\mathrm{Mod}_{S^{hL}}$ of the relative indexed tensors for these orbits. After extension to S, the orbit of gH contributes the factors indexed by L/H_g. Regrouping these factors over all double cosets is the base-change equivalence of Construction 6.5. Since both sides carry the same semilinear L-action, they descend to the same comparison in the L-fiber.

The same calculation applies to mixed inputs. Let $\mathcal{C} \to \mathcal{D} \to \mathcal{Q}$ be an S^{hH}-linear diagram. Choose an input $\mathcal{X}_{kH}$ for each $kH \in K/H$, and let $L \leq K$ be a subgroup that preserves the choices. The orbit decomposition then defines the relative indexed tensor in the L-fiber, and its scalar extension to S is

$$\bigotimes_{kH \in K/H,\, S} (\mathcal{X}_{kH} \otimes_{S^{hH},k} S),$$

with L-action by transport and permutation. Let $J \subseteq K/H$ specify the positions of one input, and let K_J be its stabilizer. By naturality of balancing, we obtain a comparison of the mixed coefficient tensor with the relative indexed tensor in every fiber $L \leq K_J$. In particular, this identifies the mixed vertices of the cube and the mixed tensors with quotient factors.

A face $J \subseteq J'$ is defined in every fiber $L \leq K_J \cap K_{J'}$, and on the cover it is the tensor product of the input maps in the changed positions. Since balancing is natural in functors and natural transformations, the associativity and base-change coherences identify all faces and higher simplices. By semilinear descent, the full relative indexed cube, with its subgroup fibers and face maps, is therefore identified with the cube of coefficient tensors.

6.2 Relative motives and motivic localization

Definition 6.6 (Relative motivic equivalences)**.** Let T be a commutative ring spectrum. A T-linear exact functor $F : \mathcal{A} \to \mathcal{B}$ between small perfect T-linear categories is a *relative motivic equivalence* if $U_T(F)$ is an equivalence in $\mathrm{Mot}_T^{\mathrm{loc}}$, where

$$U_T : \mathrm{Cat}_T^{\mathrm{perf}} \longrightarrow \mathrm{Mot}_T^{\mathrm{loc}}$$

is the universal finitary localizing invariant.

Recall that a functor $L : \mathcal{A} \to \mathcal{B}$ is a *Dwyer–Kan localization* at a class W of morphisms if, for every ∞-category $\mathcal{E}$, precomposition induces a fully faithful functor

$$L^* : \mathrm{Fun}(\mathcal{B}, \mathcal{E}) \longrightarrow \mathrm{Fun}(\mathcal{A}, \mathcal{E})$$

whose essential image consists of the functors that invert W.

Since the norms $\mathrm{N}_{H,S/R}^K$ are not exact, we need the next proposition: it lets us descend functors that preserve relative motivic equivalences, together with natural transformations and their coherences. Its proof uses the relative compactly generated presentation of [18, Theorem 1.37], which builds on [19, Appendix B].

Proposition 6.7 (Relative Dwyer–Kan presentation)**.** *Let T be a commutative ring spectrum, and let*

$$U_T : \mathrm{Cat}_T^{\mathrm{perf}} \longrightarrow \mathrm{Mot}_T^{\mathrm{loc}}$$

be the universal finitary localizing invariant. Then U_T is a Dwyer–Kan localization at the class of T-linear functors that U_T inverts. Consequently, for every presentable stable category

$\mathcal{E}$ and every $r \geq 1$, precomposition with $U_T^{\times r}$ is fully faithful from functors $\prod_1^r \mathrm{Mot}_T^{\mathrm{loc}} \to \mathcal{E}$ to functors $\prod_1^r \mathrm{Cat}_T^{\mathrm{perf}} \to \mathcal{E}$, and its essential image on separately colimit-preserving functors consists of the functors that are finitary and localizing in each variable.

Proof. Let $\mathcal{V}$ be a locally rigid compactly generated symmetric monoidal presentable stable category, and $\mathrm{Mod}_{\mathcal{V},\omega}$ the category of compactly generated $\mathcal{V}$-modules and compact-preserving $\mathcal{V}$-linear functors. By [19, Theorem B.6], in the form of [18, Theorem 1.37], the functor $\mathrm{Mod}_{\mathcal{V},\omega} \to \mathrm{Mot}_{\mathcal{V}}^{\mathrm{loc}}$ is a Dwyer–Kan localization, and the fully faithful functors are the cofibrations of a cofibration structure on the source. We take $\mathcal{V} = \mathrm{Mod}_T$. Passing to compact objects identifies $\mathrm{Mod}_{\mathrm{Mod}_T,\omega}$ with $\mathrm{Cat}_T^{\mathrm{perf}}$ and the functor with U_T. By [6, Theorem 1.19], the motives defined through compactly generated Mod_T-modules agree with $\mathrm{Mot}_T^{\mathrm{loc}}$, defined through dualizable ones. This proves the first assertion.

For r variables we apply the first assertion one variable at a time, using the equivalence $\mathrm{Fun}(\mathcal{A} \times \mathcal{B}, \mathcal{E}) \simeq \mathrm{Fun}(\mathcal{A}, \mathrm{Fun}(\mathcal{B}, \mathcal{E}))$. Applying the universal property of $\mathrm{Mot}_T^{\mathrm{loc}}$ in each variable, we then identify separately colimit-preserving functors out of $\prod_1^r \mathrm{Mot}_T^{\mathrm{loc}}$ with functors that are finitary and localizing in each variable. □

A genuine functor in the sense of Recollections 5.1 is *semiadditive* if it also commutes with additive transfers. It is *fiberwise finitary and localizing* if each component $(\underline{\mathrm{Cat}}_{S/R}^{\mathrm{perf}})^H \to \mathcal{E}^H$ is a finitary localizing invariant. We write $\underline{\mathrm{Mot}}_{S/R}^{\mathrm{loc}}$ for the assignment $G/H \mapsto \mathrm{Mot}_{S^{hH}}^{\mathrm{loc}}$.

Proposition 6.8 (Relative genuine motives)**.** *The functors $U_{S^{hH}}$ assemble to a semiadditive genuine functor $\underline{U}_{S/R} : \underline{\mathrm{Cat}}_{S/R}^{\mathrm{perf}} \to \underline{\mathrm{Mot}}_{S/R}^{\mathrm{loc}}$, where*

$$\underline{\mathrm{Cat}}_{S/R}^{\mathrm{perf}}\left(\coprod_i G/H_i\right) = \prod_i \mathrm{Cat}_{S^{hH_i}}^{\mathrm{perf}},$$
$$(\underline{\mathrm{Mot}}_{S/R}^{\mathrm{loc}})^H = \mathrm{Mot}_{S^{hH}}^{\mathrm{loc}}.$$

The target is the genuine G-category of relative localizing motives associated with S/R. *It is G-presentable and G-stable. Its restrictions are extension of coefficients and its additive transfers are restriction of coefficients; both are strongly continuous. For any G-presentable stable category $\mathcal{E}$, precomposition with $\underline{U}_{S/R}$ identifies G-colimit-preserving functors $\underline{\mathrm{Mot}}_{S/R}^{\mathrm{loc}} \to \mathcal{E}$ with semiadditive genuine functors $\underline{\mathrm{Cat}}_{S/R}^{\mathrm{perf}} \to \mathcal{E}$ that are fiberwise finitary and localizing. This description commutes with restriction to subgroups.*

Proof. Since extension and restriction of coefficients preserve Karoubi sequences, finite sums and filtered colimits, they extend to colimit-preserving functors on motives. By Proposition 6.7, their adjunctions, composition maps and Beck–Chevalley equivalences descend along $\underline{U}_{S/R}$. On the splitting cover, the Beck–Chevalley maps only reindex a product of copies of S by double cosets. Thus both adjunctions in (6.1) hold on motives. Since the adjoints preserve colimits, both functors are strongly continuous.

The fibers are presentable and stable, and the restrictions preserve colimits and have the left adjoints above. Hence $\underline{\mathrm{Mot}}_{S/R}^{\mathrm{loc}}$ has all genuine colimits [10, Recollections 2.1.3] and is G-presentable [10, Recollections 2.1.5]. The genuine category $\underline{\mathrm{Cat}}_{S/R}^{\mathrm{perf}}$ is G-semiadditive, and its finite indexed coproducts preserve motivic equivalences, since they are built from finite sums and restriction of coefficients. Therefore the fiberwise Dwyer–Kan localization $\underline{\mathrm{Mot}}_{S/R}^{\mathrm{loc}}$ is G-semiadditive [10, Lemma 2.1.23], hence G-stable [10, Definition 2.1.10].

For the universal property, we extend a given invariant fiberwise by the universal property of relative localizing motives. By Proposition 6.7, its compatibilities with restrictions and

transfers descend to the extension. The extension preserves fiberwise colimits and finite indexed coproducts, and hence all genuine colimits. □

Theorem 6.9 (Genuine symmetric monoidal localization)**.** *Every structure norm* $\mathrm{N}^K_{H,S/R}$ *on* $\underline{\mathrm{Cat}}^{\mathrm{perf}}_{S/R}$ *preserves relative motivic equivalences. Consequently the fiberwise Dwyer–Kan localization*

$$\underline{U}_{S/R} : \underline{\mathrm{Cat}}^{\mathrm{perf}}_{S/R} \longrightarrow \underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R}$$

induces a genuine G-symmetric monoidal structure on $\underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R}$ *for which* $\underline{U}_{S/R}$ *is genuine symmetric monoidal. The target is G-presentable and G-stable. Its restrictions are extension of coefficients and its additive transfers are restriction of coefficients; both are strongly continuous.*

We also write $\mathrm{N}^K_{H,S/R} : (\underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R})^H \to (\underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R})^K$ for the descended norm. Since the localization is genuine symmetric monoidal, we obtain the commutative square

$$\begin{array}{ccc} (\underline{\mathrm{Cat}}^{\mathrm{perf}}_{S/R})^H & \xrightarrow{\mathrm{N}^K_{H,S/R}} & (\underline{\mathrm{Cat}}^{\mathrm{perf}}_{S/R})^K \\ {\scriptstyle U_{S^{hH}}}\downarrow & & \downarrow{\scriptstyle U_{S^{hK}}} \\ (\underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R})^H & \xrightarrow{\mathrm{N}^K_{H,S/R}} & (\underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R})^K, \end{array}$$

compatibly with composition, units and base change.

6.3 Norms preserve motivic equivalences

Since $\underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R}$ is genuine presentable stable (Proposition 6.8), we have, for each $K \leq G$, the geometric quotient of Construction 5.3,

$$q_{K,S/R} : \mathrm{Mot}^{\mathrm{loc}}_{S^{hK}} \longrightarrow (\underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R})^{K,\mathrm{geom}},$$

and Lemma 5.4 applies to it. In this setting the transfer $\mathrm{Ind}^K_J : \mathrm{Mot}^{\mathrm{loc}}_{S^{hJ}} \to \mathrm{Mot}^{\mathrm{loc}}_{S^{hK}}$ is restriction of coefficients along $S^{hK} \to S^{hJ}$, and we use the same notation for linear categories.

Transfers and tensor products satisfy the projection formula

$$(\mathrm{Ind}^K_J \, \mathcal{E}) \otimes_{S^{hK}} \mathcal{D} \simeq \mathrm{Ind}^K_J (\mathcal{E} \otimes_{S^{hJ}} \mathrm{res}^K_J \, \mathcal{D}). \tag{6.3}$$

For coefficient categories, this follows from semilinear descent, since on the finite étale splitting cover both sides are the same finite product of relative tensor products over S, with the same semilinear action. It also holds for motives. After we apply $U_{S^{hK}}$, both sides are finitary and localizing in each coefficient variable, because relative tensor products, restriction and additive transfer have these properties. The relative localizing universal property therefore extends them to separately colimit-preserving functors on $\mathrm{Mot}^{\mathrm{loc}}_{S^{hJ}} \times \mathrm{Mot}^{\mathrm{loc}}_{S^{hK}}$, and by Proposition 6.7 the equivalence extends to these functors. Since the kernel of $q_{K,S/R}$ is generated as a localizing subcategory by proper transfers, the projection formula shows that this kernel is a tensor ideal. It follows that $q_{K,S/R}$ is symmetric monoidal.

By a coherent descent of the norms to $\underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R}$ we mean norms $\mathrm{N}^K_{H,S/R}$ on motives, with all their coherences, for which $\underline{U}_{S/R}$ is genuine symmetric monoidal. Given such norms and subgroups $H_1, \ldots, H_r \leq K$, the corresponding geometric composite is

$$\prod_a \mathrm{Mot}^{\mathrm{loc}}_{S^{hH_a}} \longrightarrow (\underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R})^{K,\mathrm{geom}}, \qquad (M_a)_a \longmapsto q_{K,S/R}\left(\bigotimes_a \mathrm{N}^K_{H_a,S/R} M_a\right).$$

Proposition 6.10 (Motivic descent of norms)**.** *The norms* $\mathrm{N}^K_{H,S/R}$ *preserve relative motivic equivalences and descend coherently to norms on* $\underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R}$*. Their geometric composites factor as separately colimit-preserving functors*

$$\prod_{a=1}^{r} (\underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R})^{H_a,\mathrm{geom}} \longrightarrow (\underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R})^{K,\mathrm{geom}}. \tag{6.4}$$

A norm $\mathrm{N}^K_{H,S/R}$ need not preserve Karoubi sequences, but the failure is concentrated in categories induced from proper subgroups. We therefore filter the norm of a Karoubi sequence by the number of quotient factors (Corollary 6.11). Since the intermediate layers are properly induced, the norm becomes exact after $q_{K,S/R}$ (Lemma 6.12). On proper subgroups we argue by induction on $|K|$, and isotropy separation combines the two.

Applied to coefficient categories, Proposition 5.15 gives the following filtration.

Corollary 6.11 (Finite étale descent of tensor filtrations)**.** *For a Karoubi sequence* $\mathcal{A} \hookrightarrow \mathcal{B} \to \mathcal{C}$ *over* S^{hH}*, put* $d = |K/H|$*, and write* K_J *for the stabilizer of a subset* $J \subseteq K/H$*. There is a filtration by fully faithful maps*

$$\mathrm{N}^K_{H,S/R}\mathcal{A} = L_0 \hookrightarrow L_1 \hookrightarrow \cdots \hookrightarrow L_d = \mathrm{N}^K_{H,S/R}\mathcal{B}$$

with $L_d/L_{d-1} \simeq \mathrm{N}^K_{H,S/R}\mathcal{C}$*. For* $J \subseteq K/H$*, let* $\mathcal{E}_J$ *be the* S^{hK_J}*-linear descent of the mixed tensor with quotient factors in* J *and* $\mathcal{A}$*-factors outside* J*. The intermediate layers are*

$$\mathrm{cofib}(L_{j-1} \to L_j) \simeq \bigoplus_{[J]\in K\backslash\{J\subseteq K/H : |J|=j\}} \mathrm{Ind}^K_{K_J} \mathcal{E}_J, \qquad 0 < j < d, \tag{6.5}$$

where each displayed K_J *is proper in* K*. These assertions hold for small perfect categories and for dualizable linear categories with strongly continuous maps. In the latter case, the filtration maps are strongly continuous and the skeletal colimits agree with those in presentable categories.*

Proof. We apply Proposition 5.15 to the genuine category of coefficients with fibers $\mathrm{Mod}_{S^{hJ}}$, with direction set K/H and input sequence $\mathcal{A} \hookrightarrow \mathcal{B} \to \mathcal{C}$. The comparison of (5.7) with $p_\otimes$ identifies the endpoints of the resulting filtration with the coefficient norms. On the splitting cover, the layer at J is

$$\left(\bigotimes_{kH\in J,\, S} (\mathcal{C} \otimes_{S^{hH},k} S)\right) \otimes_S \left(\bigotimes_{kH\notin J,\, S} (\mathcal{A} \otimes_{S^{hH},k} S)\right),$$

with its semilinear K_J-action, that is, the scalar extension of $\mathcal{E}_J$. Hence the orbit-induction formula of Proposition 5.15 becomes (6.5). The subgroups K_J occurring there are proper, since a nonempty proper subset of the transitive K-set K/H has proper stabilizer. The same proposition gives full faithfulness, strong continuity, and the comparison of skeletal colimits.

For small perfect categories, we apply the presentable assertion to Ind-completions. Relative tensor products of compact generators are compact generators, and all maps in the cube preserve compact objects. Hence the finite skeletal colimits and the quotients are compactly generated, with compact-preserving maps. These properties descend by Lemma 6.4, and passing to compact objects gives the statement for small perfect categories. □

The next lemma gives the exactness after $q_{K,S/R}$. A section σ of $K/L \to K/H$ is a map of sets, and K acts on sections by $(k\sigma)(x) = k\,\sigma(k^{-1}x)$.

Lemma 6.12 (Norms after the geometric quotient)**.** *For $H_1, \dots, H_r \leq K$, the functor*

$$(\mathcal{C}_1, \dots, \mathcal{C}_r) \longmapsto q_{K,S/R} U_{S^{hK}} \left(\bigotimes_{a=1}^{r} \mathrm{N}^K_{H_a,S/R} \mathcal{C}_a \right) \tag{6.6}$$

is finitary and localizing in each variable.

For $L < H \leq K$, let $\mathcal{C}$ be an S^{hL}-linear category. Let $(\mathcal{C}_y)_{y \in K/L}$ be its conjugates after extension to S. For a section σ of $K/L \to K/H$, write K_σ for its stabilizer and $\mathcal{D}_\sigma$ for the S^{hK_σ}-linear descent of $\bigotimes_{x \in K/H,\, S} \mathcal{C}_{\sigma(x)}$. There is a natural decomposition

$$\mathrm{N}^K_{H,S/R} \operatorname{Ind}^H_L \mathcal{C} \simeq \bigoplus_{[\sigma] \in K \backslash \operatorname{Sect}(K/L \to K/H)} \operatorname{Ind}^K_{K_\sigma} \mathcal{D}_\sigma, \qquad K_\sigma < K. \tag{6.7}$$

Consequently, its motive vanishes after $q_{K,S/R}$, also after tensoring with any fixed coefficient category.

Proof. To see that (6.6) is localizing, we apply Corollary 6.11 to a Karoubi sequence in one input and tensor the resulting filtration with the norms of the other inputs. Each step $L_{j-1} \hookrightarrow L_j$ is a Karoubi sequence, and hence $U_{S^{hK}}$ takes it to a cofiber sequence. By (6.3), the layers with $0 < j < d$ remain properly induced after tensoring, so they are killed by $q_{K,S/R}$. What remains is a cofiber sequence of the values of (6.6) at $\mathcal{A}$, $\mathcal{B}$ and $\mathcal{C}$. Filtered-colimit preservation follows from (6.2), as for the norm itself.

For (6.7), we extend scalars to S and distribute the norm over the sums. The factor at $x \in K/H$ is the sum of the $\mathcal{C}_y$ over the points y above x, so the summands are the tensor products $\bigotimes_x \mathcal{C}_{\sigma(x)}$ for sections σ. The semilinear K-action permutes these summands as it permutes sections. Therefore, as in the proof of Corollary 6.11, the sum over the orbit of σ descends to $\operatorname{Ind}^K_{K_\sigma} \mathcal{D}_\sigma$. Each K_σ is proper, since the image of a K-fixed section would be a K-stable subset of the transitive K-set K/L, hence all of K/L, which is impossible since $|K/L| > |K/H|$. The assertion after tensoring follows from (6.3). □

Proof of Proposition 6.10. By the relative localizing universal property, relative tensor products preserve motivic equivalences in each variable separately. Over an orbit G/K, a norm along a map of finite G-sets is a relative tensor product of transitive norms $\mathrm{N}^K_{H,S/R}$, one for each K-orbit of the fiber. It therefore suffices to treat $\mathrm{N}^K_{H,S/R}$, and we argue by induction on $|K|$. The case $K = e$ is the assertion for ordinary tensor products.

By Lemma 6.12, the functor $q_{K,S/R} U_{S^{hK}} \mathrm{N}^K_{H,S/R}$ is a finitary localizing invariant, and hence it inverts motivic equivalences. For $J < K$, splitting the J-set K/H into the orbits $JgH/H \cong J/(J \cap {}^gH)$, we obtain the double-coset formula

$$\operatorname{res}^K_J \mathrm{N}^K_{H,S/R} \mathcal{C} \simeq \bigotimes_{[g] \in J \backslash K/H} \mathrm{N}^J_{J \cap {}^gH,S/R} \left(\operatorname{res}^{{}^gH}_{J \cap {}^gH} ({}^g\mathcal{C}) \right). \tag{6.8}$$

Restriction and conjugation preserve motivic equivalences (Proposition 6.8), so by the induction hypothesis the right side of (6.8) preserves them. Hence, for a motivic equivalence ϕ over S^{hH}, the map $U_{S^{hK}} \mathrm{N}^K_{H,S/R} \phi$ becomes an equivalence after every res^K_J with $J < K$ and after $q_{K,S/R}$. By Lemma 5.4, it is therefore an equivalence.

It remains to descend the norms coherently. By the genuine symmetric monoidal localization principle [10, Recollections 2.1.22 and proof of Theorem 3.2.2], if the tensor products and norms of a genuine symmetric monoidal category preserve a fiberwise class of maps, then the fiberwise Dwyer–Kan localization at that class inherits a genuine symmetric monoidal structure for which the localization is genuine symmetric monoidal. Applied to motivic

equivalences, it descends the norms, and Proposition 6.7 descends their coherences. Hence $\underline{U}_{S/R} : \underline{\mathrm{Cat}}^{\mathrm{perf}}_{S/R} \to \underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R}$ is a genuine symmetric monoidal localization.

For (6.4), we apply the relative localizing universal property, one variable at a time, to the finitary localizing functor (6.6). This extends it to a separately colimit-preserving functor on $\prod_a \mathrm{Mot}^{\mathrm{loc}}_{S^{hH_a}}$, and by Proposition 6.7 the extension is the descended geometric composite. By (6.7), this functor vanishes when one coefficient input is a proper transfer, and by Proposition 6.7 the same holds for all motive inputs. Since it preserves colimits in each variable separately, it kills the localizing subcategory generated by proper transfers in each input. This gives the factorization (6.4). □

Proof of Theorem 6.9. Proposition 6.8 constructs the genuine fiberwise Dwyer–Kan localization and shows that it is presentable and stable, with strongly continuous restrictions and transfers. By Proposition 6.10, the norms descend to it together with their coherences. □

6.4 Distributivity and strong continuity

We next show that the descended structure on $\underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R}$ is distributive in the sense of Definition 5.13 (Proposition 6.15). For a genuine presentable stable G-category, the geometric quotients q_K of Construction 5.3 reduce distributivity to a condition at each subgroup.

Lemma 6.13 (Geometric quotients of indexed colimits)**.** *Let $\mathcal{C}$ be a genuine presentable stable K-category, let $q_K : \mathcal{C}^K \to \mathcal{C}^{K,\mathrm{geom}}$ be its top geometric quotient, and let X be a diagram indexed by a K-category $\mathcal{J}$. Then*

$$q_K \operatorname*{colim}^{K}_{\mathcal{J}} X \simeq \operatorname*{colim}_{\mathcal{J}^K} q_K X^K. \tag{6.9}$$

On the left, the superscript denotes the top fiber of the indexed colimit, and the colimit on the right is ordinary.

Proof. Let i_K be the fully faithful right adjoint to q_K, and let $Z \in \mathcal{C}^{K,\mathrm{geom}}$. By the description of the essential image of i_K in Construction 5.3, we have $\mathrm{res}^K_J\, i_K Z = 0$ for $J < K$. Therefore a genuine cone $X \to i_K Z$ has unique data on the proper fibers, and its remaining data form an ordinary cone $X^K \to i_K Z$ on $\mathcal{J}^K$. By the universal property of the genuine colimit and the adjunction $q_K \dashv i_K$, we obtain

$$\mathrm{Map}\left(q_K \operatorname*{colim}^{K}_{\mathcal{J}} X, Z\right) \simeq \lim_{(\mathcal{J}^K)^{\mathrm{op}}} \mathrm{Map}(q_K X^K(-), Z),$$

naturally in Z, and (6.9) follows. □

Lemma 6.14 (A geometric criterion for indexed distributivity)**.** *Let $\mathcal{C}$ be a genuine G-presentable stable category equipped with a genuine G-symmetric monoidal structure. For every $K \leq G$, let $q_K : \mathcal{C}^K \to \mathcal{C}^{K,\mathrm{geom}}$ be the geometric quotient of Construction 5.3.*

Suppose that, for every finite family $H_1, \dots, H_r \leq K$, the indexed multiplication associated with $\coprod_{a=1}^r K/H_a$ admits a natural factorization

$$\begin{array}{ccc} \displaystyle\prod_{a=1}^{r} \mathcal{C}^{H_a} & \longrightarrow & \mathcal{C}^K \\ {\scriptstyle \prod_a q_{H_a}}\downarrow & & \downarrow{\scriptstyle q_K} \\ \displaystyle\prod_{a=1}^{r} \mathcal{C}^{H_a,\mathrm{geom}} & \longrightarrow & \mathcal{C}^{K,\mathrm{geom}}, \end{array}$$

whose lower functor preserves colimits separately in each variable. The factorization includes a specified natural equivalence making the square commute. Then the given genuine symmetric monoidal structure is distributive.

Proof. We show that the indexed distributivity comparison of [10, Definition 2.1.13] is an equivalence after base change to each orbit G/K, by induction on $|K|$. The source of the map of finite K-sets is $\coprod_a K/H_a$. For each a, choose a diagram X_a indexed by an H_a-category $\mathcal{J}_a$. The comparison is the canonical map from the indexed colimit, over the indexed product of the $\mathcal{J}_a$, of the indexed multiplication of the X_a to the indexed multiplication of their indexed colimits. We show that it is an equivalence after every proper restriction and after q_K.

For $L < K$, the base-change equivalences of the genuine symmetric monoidal structure identify the restricted comparison with the comparison at L for the L-orbits of the source, which is an equivalence by induction.

By Lemma 6.13, the image under q_K of the source of the comparison is an ordinary colimit over the K-fixed fiber of the indexed product of the $\mathcal{J}_a$, which is $\prod_a \mathcal{J}_a^{H_a}$. By the assumed factorization, the diagram of this colimit is the lower functor applied to the diagrams $q_{H_a} X_a^{H_a}$.

By the same lemma, the image of each input colimit in $\mathcal{C}^{H_a,\mathrm{geom}}$ is $\operatorname{colim}_{\mathcal{J}_a^{H_a}} q_{H_a} X_a^{H_a}$. The comparison after q_K is therefore the canonical map from the colimit over $\prod_a \mathcal{J}_a^{H_a}$ of the values of the lower functor to its value on the input colimits. This map is an equivalence because the lower functor preserves colimits separately in each variable. This includes empty indexing categories, and for an empty family of inputs the comparison is the identity of the unit.

Since proper restrictions and q_K jointly detect equivalences (Lemma 5.4), the comparison is an equivalence at K. For $K = e$, the geometric quotient is the identity. □

For $f : G/H \to G/K$, we apply the presentable indexed tensor of Definition 5.5 to $\mathrm{res}^G_H \underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R}$. Its universal distributive functor is

$$\mu_{S/R,f} : f_* \,\mathrm{res}^G_H \underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R} \longrightarrow \mathrm{N}^{K,\mathrm{Pr}}_H \,\mathrm{res}^G_H \underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R},$$

where f_* is coinduction, so that $(f_* \,\mathrm{res}^G_H \underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R})^K = (\underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R})^H$. The universal property of $\mu_{S/R,f}$ is an equivalence of functor categories [10, Fact D.0.2(2)]. A model of $\mathrm{N}^{K,\mathrm{Pr}}_H$ in terms of modules is given in Appendix A.

Proposition 6.15 (Indexed distributivity)**.** *The descended genuine symmetric monoidal structure on $\underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R}$ is distributive. For $H \le K \le G$, the indexed multiplication factors through $\mu_{S/R,f}$ by a genuine colimit-preserving symmetric monoidal functor*

$$m^K_{H,S/R} : \mathrm{N}^{K,\mathrm{Pr}}_H(\mathrm{res}^G_H \underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R}) \longrightarrow \mathrm{res}^G_K \underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R}. \tag{6.10}$$

Proof. By Proposition 6.10, the geometric composites of the descended indexed multiplication factor through the separately colimit-preserving functors (6.4). Hence the structure is distributive by Lemma 6.14.

By the universal property [10, Fact D.0.2(2)], the multiplication then factors through $\mu_{S/R,f}$ by a colimit-preserving functor $m^K_{H,S/R}$. The natural equivalence in the norm square following Theorem 6.9 comes from Proposition 6.10.

A distributive genuine symmetric monoidal structure is a commutative algebra in genuine presentable categories [10, Example 2.1.15]. Hence the indexed multiplication of $\underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R}$ is a map of commutative algebras $\mathrm{N}^{K,\mathrm{Pr}}_H(\mathrm{res}^G_H \underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R}) \to \mathrm{res}^G_K \underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R}$, where the algebra

structure on the source comes from the symmetric monoidal 2-functor $\mathrm{N}_H^{K,\mathrm{Pr}}$ [10, Proposition D.0.4]. We apply the functor-category equivalence of [10, Fact D.0.2(2)] to natural transformations in each variable. It identifies the underlying functor of this map with $m^K_{H,S/R}$, and its multiplication and unit comparisons with those of the indexed multiplication. On category motives, these are the comparisons for the coefficient norms, descended by Proposition 6.7. Thus $m^K_{H,S/R}$ is symmetric monoidal. □

We call $m^K_{H,S/R}$ the *linearized multiplication* associated with S/R and $G/H \to G/K$. Together with its factorization through $\mu_{S/R,f}$, it is unique up to contractible choice by [10, Fact D.0.2(2)].

The norm $(\underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R})^H \to (\underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R})^K$ need not be exact, so we apply continuous K-theory to the linearized multiplication instead, once we have shown that it is strongly continuous.

Under the identification $(f_* \operatorname{res}^G_H \underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R})^K = (\underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R})^H$, the norm is the composite

$$(\underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R})^H \xrightarrow{\mu^K_{S/R,f}} (\mathrm{N}_H^{K,\mathrm{Pr}} \operatorname{res}^G_H \underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R})^K \xrightarrow{(m^K_{H,S/R})^K} (\underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R})^K,$$

where the superscripts K denote evaluation at G/K. With $p : S^{hK} \to S^{hH}$, we have, for every small perfect S^{hH}-linear category $\mathcal{A}$,

$$(m^K_{H,S/R})^K(\mu^K_{S/R,f}(U_{S^{hH}}\mathcal{A})) \simeq U_{S^{hK}}(p_\otimes \mathcal{A}).$$

Proposition 6.16 (Strong continuity of the multiplication)**.** *The source and target of every $m^K_{H,S/R}$ in (6.10) are rigid, and hence dualizable, over genuine K-spectra, and $m^K_{H,S/R}$ is strongly continuous.*

Proof. For every $J \leq G$, the base $\mathrm{Mod}_{S^{hJ}}$ is rigid symmetric monoidal, so $\mathrm{Mot}^{\mathrm{loc}}_{S^{hJ}}$ is rigid over Sp by Efimov's theorem [7, Theorem 3.1(ii)]. In particular its unit is compact [7, Section 1.2]. By the equivalence

$$\mathrm{Pr}^L_{J,\mathrm{st}} \simeq \mathrm{Mod}_{\mathrm{Sp}^J}(\mathrm{Pr}^L_{\mathrm{st}})$$

of [10, Theorem B.0.1], applied to the genuine symmetric monoidal structure of Proposition 6.15, the category $\mathrm{Mot}^{\mathrm{loc}}_{S^{hJ}}$ is a commutative Sp^J-algebra. The category Sp^J is rigid over Sp, since its compact orbit generators are dualizable, and so is $\mathrm{Mot}^{\mathrm{loc}}_{S^{hJ}}$. By the relative rigidity criterion [17, Corollary 4.46], it follows that $\mathrm{Mot}^{\mathrm{loc}}_{S^{hJ}}$ is rigid over Sp^J. By Definition 5.14, rigidity over Sp^J implies that $\operatorname{res}^G_J \underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R}$ is dualizable over Sp^J, as is needed to apply K^{cont}_J.

For the source, we use that $\mathrm{N}_H^{K,\mathrm{Pr}}$ is a symmetric monoidal 2-functor on genuine presentable categories [10, Proposition D.0.4]. Since the right adjoints in the rigidity data preserve colimits, $\mathrm{N}_H^{K,\mathrm{Pr}}$ transports these adjunctions with their triangle identities and the bimodule linearity of the right adjoint to the multiplication. As $\mathrm{N}_H^{K,\mathrm{Pr}}$ carries Sp^H to Sp^K, it follows that rigidity of $\operatorname{res}^G_H \underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R}$ over Sp^H implies rigidity of $\mathrm{N}_H^{K,\mathrm{Pr}} \operatorname{res}^G_H \underline{\mathrm{Mot}}^{\mathrm{loc}}_{S/R}$ over Sp^K. The same holds after every subgroup restriction.

The functor $m^K_{H,S/R}$ is symmetric monoidal and colimit-preserving (Proposition 6.15). Its source is rigid, and in particular locally rigid, over Sp^K. The unit of its target is Sp^K-atomic. Indeed, the Sp^K-valued mapping functor out of the unit is the right adjoint of the unit functor $\mathrm{Sp}^K \to \mathrm{Mot}^{\mathrm{loc}}_{S^{hK}}$, and by rigidity of the target this right adjoint is Sp^K-linear and preserves colimits. By the relative adjoint criterion [17, Observation 4.51], the functor $m^K_{H,S/R}$ therefore has a colimit-preserving Sp^K-linear right adjoint. Hence $m^K_{H,S/R}$ is strongly continuous. □

6.5 Finite étale invariance

We now prove the remaining implication of Theorem 2.10. By Lemma 2.3, it suffices to factor the unit of $K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}})$ through an E-acyclic spectrum. For a G-torsor S/R, we use Theorem 5.6 and Proposition 6.16 to factor this unit through the categorical fixed points of $\mathrm{N}_e^G K^{\mathrm{cont}}(\mathrm{Mot}_S^{\mathrm{loc}})$. The next lemma shows that these fixed points are E-acyclic.

Lemma 6.17 (Acyclicity of categorical fixed points of a norm)**.** *If $E \otimes X = 0$ for spectra E, X, then $E \otimes (\mathrm{N}_e^G X)^H = 0$ for every $H \leq G$. Here $(-)^H$ denotes genuine categorical fixed points.*

Proof. Regard E as $E \otimes \mathbb{S}_G$. For every $J \leq G$,

$$\Phi^J\big((E \otimes \mathbb{S}_G) \otimes \mathrm{N}_e^G X\big) \simeq E \otimes X^{\otimes |G/J|} = 0,$$

so $(E \otimes \mathbb{S}_G) \otimes \mathrm{N}_e^G X = 0$, since geometric fixed points are jointly conservative. Categorical fixed points preserve colimits and are therefore Sp-linear. Hence

$$E \otimes (\mathrm{N}_e^G X)^H \simeq \big((E \otimes \mathbb{S}_G) \otimes \mathrm{N}_e^G X\big)^H = 0.$$

□

To factor the unit, we apply Theorem 5.6 to $\underline{\mathrm{Mot}}_{S/R}^{\mathrm{loc}}$. Its fixed points satisfy

$$\big(K_G^{\mathrm{cont}}(\underline{\mathrm{Mot}}_{S/R}^{\mathrm{loc}})\big)^H \simeq K^{\mathrm{cont}}(\mathrm{Mot}_{S^{hH}}^{\mathrm{loc}}),$$

The linearized multiplication for $f : G/e \to G/G$ is strongly continuous by Proposition 6.16, and applying K_G^{cont} to it gives

$$\begin{aligned} \mathrm{N}_e^G K^{\mathrm{cont}}(\mathrm{Mot}_S^{\mathrm{loc}}) &\longrightarrow K_G^{\mathrm{cont}}(\mathrm{N}_e^{G,\mathrm{Pr}} \operatorname{res}_e^G \underline{\mathrm{Mot}}_{S/R}^{\mathrm{loc}}) \\ &\xrightarrow{K_G^{\mathrm{cont}}(m_{e,S/R}^G)} K_G^{\mathrm{cont}}(\underline{\mathrm{Mot}}_{S/R}^{\mathrm{loc}}). \end{aligned} \tag{6.11}$$

Both arrows are compatible with the unit maps from $\mathbb{S}_G$. For the first, this is the last assertion of Theorem 5.6, applied to $\mathrm{Mot}_S^{\mathrm{loc}}$, whose unit is compact. For the second, $m_{e,S/R}^G$ is symmetric monoidal. Hence the unit $\mathbb{S}_G \to K_G^{\mathrm{cont}}(\underline{\mathrm{Mot}}_{S/R}^{\mathrm{loc}})$ factors through $\mathrm{N}_e^G K^{\mathrm{cont}}(\mathrm{Mot}_S^{\mathrm{loc}})$.

Proof of Theorem 2.10. Fix a spectrum E. Suppose first that S/R is a G-torsor and that $E \otimes K^{\mathrm{cont}}(\mathrm{Mot}_S^{\mathrm{loc}}) = 0$. We take categorical G-fixed points of the factorization of $\mathbb{S}_G \to K_G^{\mathrm{cont}}(\underline{\mathrm{Mot}}_{S/R}^{\mathrm{loc}})$ through (6.11), and we precompose with the unit $\mathbb{S} \to (\mathbb{S}_G)^G$. Since categorical fixed points are lax symmetric monoidal, the composite is the unit of $(K_G^{\mathrm{cont}}(\underline{\mathrm{Mot}}_{S/R}^{\mathrm{loc}}))^G \simeq K^{\mathrm{cont}}(\mathrm{Mot}_{S^{hG}}^{\mathrm{loc}}) = K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}})$. It factors through $(\mathrm{N}_e^G K^{\mathrm{cont}}(\mathrm{Mot}_S^{\mathrm{loc}}))^G$, which is E-acyclic by Lemma 6.17. Therefore $E \otimes K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}}) = 0$ by Lemma 2.3.

Suppose next that S/R has constant rank $d > 0$. We reduce to the torsor case by means of the frame torsor. For the finite étale extension $\pi_0 R \to \pi_0 S$, let $\operatorname{Spec} T_0$ be the complement of all diagonals in the d-fold fiber product of $\operatorname{Spec} \pi_0 S$ over $\operatorname{Spec} \pi_0 R$. With Σ_d permuting the factors, T_0 is a faithful finite étale Σ_d-torsor over $\pi_0 R$, and the projection to the first factor gives $\pi_0 S \to T_0$. Since finite étale algebras over R and over $\pi_0 R$ are equivalent [15, Theorem 7.5.0.6], these lift uniquely to a faithful finite étale Σ_d-torsor T/R with an R-algebra map $S \to T$. Since the map $K^{\mathrm{cont}}(\mathrm{Mot}_S^{\mathrm{loc}}) \to K^{\mathrm{cont}}(\mathrm{Mot}_T^{\mathrm{loc}})$ is unital, $K^{\mathrm{cont}}(\mathrm{Mot}_T^{\mathrm{loc}})$ is E-acyclic by Lemma 2.3, and the torsor case for T/R gives the claim for $K^{\mathrm{cont}}(\mathrm{Mot}_R^{\mathrm{loc}})$.

In general, R splits into finitely many clopen components on which S has constant rank, and by faithfulness this rank is positive on nonzero components. Using central idempotents and the relative universal property, we obtain

$$\mathrm{Mot}^{\mathrm{loc}}_{\prod_i R_i} \simeq \prod_i \mathrm{Mot}^{\mathrm{loc}}_{R_i}, \qquad K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_{\prod_i R_i}) \simeq \prod_i K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_{R_i}),$$

compatibly with units and multiplication. Hence the constant-rank case applies to each component. The implication from R to S follows from the unital map $K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_R) \to K^{\mathrm{cont}}(\mathrm{Mot}^{\mathrm{loc}}_S)$. □

The theorem applies in particular to finite separable field extensions and their spectral base changes, such as $Hk[u^{\pm 1}] \to Hk'[u^{\pm 1}]$ for k'/k finite separable in characteristic zero and $|u|$ even.

A Indexed tensors in the module model

Let $H \leq G$ be finite groups, let $f : G/H \to G/G$ be the orbit map, and let $\mathcal{C}$ be a genuine H-presentable stable category. By [10, Theorem B.0.1], the top fiber $\mathcal{C}^H$ is a Sp^H-module, and for every $J \leq H$ we have

$$\mathcal{C}^J \simeq \mathrm{Sp}^J \otimes_{\mathrm{Sp}^H} \mathcal{C}^H,$$

where the tensor product is extension of scalars along the symmetric monoidal restriction functor $\mathrm{Sp}^H \to \mathrm{Sp}^J$. Indeed, the coinduction to H of a genuine J-presentable stable category $\mathcal{D}$ has top fiber $\mathcal{D}^J$, on which Sp^H acts through restriction. Hence coinduction is restriction of scalars, and its left adjoint, genuine restriction, is extension of scalars.

Construction A.1 (Indexed tensors in the module model)**.** We construct the presentable indexed tensor $\mathrm{N}_H^{G,\mathrm{Pr}}\mathcal{C}$ of Definition 5.5 by a genuine cocompletion followed by genuine stabilization. Regard $\mathcal{C}$ as a genuine presentable H-category, and choose a regular cardinal κ such that $\mathcal{C} \simeq \mathrm{Ind}_\kappa(\mathcal{A})$ for the small genuine H-category $\mathcal{A} = \mathcal{C}^\kappa$ of κ-compact objects. The category $\mathcal{A}$ is closed under genuine κ-small colimits [8, proof of Proposition 3.72]. Since the groups are finite, every regular cardinal is normable for finite-index norms [8, Remark 4.26].

Next we form the genuine indexed product $f_*\mathcal{A}$. For every κ-small colimit cocone $D : \mathcal{I}^\triangleright \to \mathcal{A}$, we distinguish the induced cocone

$$(f_*\mathcal{I})^\triangleright \longrightarrow f_*(\mathcal{I}^\triangleright) \xrightarrow{f_*D} f_*\mathcal{A},$$

whose first arrow is induced by the inclusion $\mathcal{I} \to \mathcal{I}^\triangleright$ and the cone point. We do the same after every base change of f. The resulting class of cocones is the indexed product of cocone classes [8, Construction 3.5].

Since κ is normable, genuine indexed products of κ-small indexing categories are again κ-small [8, Definitions 3.47 and 3.62]. Moreover, the class of colimit cocones contains the constant cocones over a terminal category. It is therefore projectable [8, Example 3.20], and by [8, Lemma 3.49] its indexed products are projectable as well.

We now freely adjoin genuine κ-small colimits to $f_*\mathcal{A}$ so that the distinguished cocones become colimit cocones. Functors out of the result that preserve genuine κ-small colimits are then exactly the distributive functors out of $f_*\mathcal{A}$. Explicitly, let y be the Yoneda embedding into genuine parametrized presheaves on $f_*\mathcal{A}$ valued in genuine G-spaces. For every subgroup and every distinguished cocone $\delta : \mathcal{J}^\triangleright \to f_*\mathcal{A}$ over it, we invert the assembly map

$$\operatorname*{colim}_{j \in \mathcal{J}} y(\delta(j)) \longrightarrow y(\delta(\infty)).$$

We then close the localized representables under genuine κ-small colimits and idempotent-complete. This is the relative cocompletion of [8, Construction 3.42 and Theorem 3.50]. By [8, Corollary 3.70], applying Ind_κ gives the corresponding presentable category. It follows from [8, Theorems 3.65 and 3.75] that this construction respects indexed tensors and has the presentable distributive universal property. In particular, it is independent of the chosen presentation.

Finally, we apply *genuine G-stabilization*. This is the smashing localization of genuine G-presentable categories onto genuine G-presentable stable ones, given by tensoring with the idempotent commutative algebra whose fiber at G/J is Sp^J. By [8, Corollary 4.7, Theorem 4.22, and Remark 4.26], it is compatible with indexed tensors. The result is $\mathrm{N}_H^{G,\mathrm{Pr}}\mathcal{C}$. Its top fiber, with its Sp^G-action, is the output in the module model.

It remains to check the universal property. Let $\mathcal{D}$ be a genuine G-presentable stable category. The stabilization adjunction and the relative cocompletion theorem give an equivalence

$$\mathrm{Fun}_G^L(\mathrm{N}_H^{G,\mathrm{Pr}}\mathcal{C},\mathcal{D}) \simeq \mathrm{Fun}_G^{f\text{-distributive}}(f_*\mathcal{C},\mathcal{D}),$$

between genuine colimit-preserving functors and the full subcategory of distributive functors. This is the universal property required in Definition 5.5. The normed cocompletion and the stabilization are both compatible with composition and base change, and by [10, Proposition D.0.4] these coherences refine to a symmetric monoidal 2-functor. In the module model, we have $\mathrm{N}_H^{G,\mathrm{Pr}}\mathrm{Sp}^H \simeq \mathrm{Sp}^G$.

For $|G/H| > 1$, the functor $\mathrm{N}_H^{G,\mathrm{Pr}}$ is not an extension of scalars along the spectral norm $\mathrm{N}_H^G : \mathrm{Sp}^H \to \mathrm{Sp}^G$. Indeed, this norm does not preserve coproducts, and hence it is not a map of commutative algebras in $\mathrm{Pr}^L_{\mathrm{st}}$. For $H \leq K \leq G$ and $f : G/H \to G/K$, the same construction over G/K yields $\mathrm{N}_H^{K,\mathrm{Pr}}\mathcal{C}$.

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